

Explicit Finite-Order Lower Bounds for High-Order Robust Composite Z Rotations

Dytchem

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Abstract

We study a minimal control problem motivated by robust composite-pulse synthesis: a target single-qubit Z rotation is realized from an imperfect Z evolution whose angle carries a common unknown multiplicative error, assisted by exact, instantaneous, and cost-free X rotations. Our main results are *explicit* lower bounds, valid at every order N , on the total Z -evolution cost T . First, an elementary bound: for all $0 < \phi \leq \pi$, $T \geq 2 [2(2N+2)! \sin^2(\phi/4)]^{1/(2N+2)}$, a cost of $4/e \approx 1.4715$ per order; it follows from a doubling of flatness—order- N matrix robustness forces the scalar carrier $h(\lambda) = 1 - \frac{1}{2} \operatorname{Re} \operatorname{Tr}(U_1^\dagger U_\lambda)$ to vanish to order $q = 2N + 2$ —combined with a single Taylor–Lagrange step at the endpoint. Second, a sharpened bound that uses the whole frequency band of the carrier: for every $a \geq 2$ and every $N \geq N_0(a)$,

$$T > 4(N+1) \left[1 - a \left(\frac{\log(2N+2)}{2N+2} \right)^{2/3} \right],$$

with an explicit threshold, concretely $a = 2.3$ with $N_0 = 5010$; the coefficient therefore approaches 4, i.e. e times the elementary one. Taking the larger of the two at each order gives a single lower bound valid for every $N \geq 1$, and the asymptotic statement $\liminf_{N \rightarrow \infty} T_{\min}(N, \phi)/N \geq 4$ follows. Both results, with all constants explicit, are machine-checked in Lean 4 with Mathlib. Numerically, at $\phi = \pi$ a local search over free amplitudes yields sequences whose costs grow by about 5.2 per order for $N \leq 8$, below the equiangular cost $(2N+1)\pi$; these are upper bounds only. We close by spelling out what the bounds rule out and where incoherent errors place the useful order.

Keywords: robust quantum control; composite pulses; systematic error; linear lower bound; Fourier approximation; formal verification; Lean.

1 Introduction

Composite-pulse methods provide a general way to suppress systematic control errors by replacing a single imperfect operation with a structured sequence whose error terms cancel order by order. Arbitrarily high-order cancellation was established in early work on composite pulses, and the optimal asymptotic scaling of the number of primitive pulses was later characterized for important pulse-length-error models [1, 2, 3]. A separate line of work has addressed lower bounds on the physical operation time of robust composite gates, including geometric bounds for first-order pulse-length-error robustness [7] and, more recently, time lower bounds for first-order robustness under bounded control amplitude [8, 9]. Order- N cancellation is the vanishing of the first N error moments, the condition that also underlies optimized dynamical decoupling and the filter-function description of open-loop control [4, 5, 6]; Section 10 returns to this comparison.

The resource model considered here is deliberately different from these standard formulations. Suppose that one available interaction generates a Z rotation but its strength is not known exactly. A second control can rotate the system about an orthogonal X axis with sufficiently high precision that, in the idealized limit, its duration may be regarded as negligible. The natural question is then not how many control pulses are needed, but how much of the imperfect interaction must unavoidably be accumulated in order to cancel the common systematic error to a specified order. This distinction matters: a lower bound on the number of primitive rotations does not immediately give a lower bound on a time resource when some rotations are assigned zero cost.

The setting also occurs naturally in Rydberg-atom control. In a recently supplied QSP-based proposal, the interaction between two Rydberg atoms is treated as an unknown but deterministic phase input, while fast Rydberg transitions are used as interleaved controls; systematic variation of the interaction strength is then suppressed by composite control [20], an as yet unpublished manuscript. Systematic control deviations are, independently of that proposal, among the dominant error sources of neutral-atom two-qubit gates [15, 16]. The present work does not analyze that two-atom implementation. Instead, it isolates a single-qubit mathematical core: what is the unavoidable amount of an uncertain Z evolution if exact transverse rotations are free?

The results proved here are explicit, finite-order statements: a lower bound on T that holds at every prescribed order N , with all constants written out. They come in two layers, and the sharper of the two is the main theorem of the paper. First, an *elementary* bound (Theorem 2.2): order- N matrix robustness forces the scalar error carrier $h(\lambda) = 1 - \frac{1}{2} \operatorname{Re} \operatorname{Tr}(U_1^\dagger U_\lambda)$ to vanish to order $q = 2N + 2$ at $\lambda = 1$, and h has bounded derivatives; one Taylor–Lagrange step at the endpoint $\lambda = 0$, where the propagator degenerates to a pure X rotation, then gives $T \geq 2[2q! \sin^2(\phi/4)]^{1/q}$. The coefficient of N is $4/e \approx 1.4715$ as $N \rightarrow \infty$, and the bound is strictly stronger than the closed form $4(N+1)/e$ at every order. Second, a *sharpened* bound (Theorem 2.3): the entire frequency band of the carrier is used, not just one Taylor remainder point. The vanishing derivatives become q vanishing moments, into which any polynomial of degree $q - 1$ may be inserted; approximating $e^{-i\mu}$ on the whole band and smoothing with a compactly supported Fourier multiplier makes $h(0)$ exponentially small whenever $\tau \leq \rho q$ with $\rho < 1$, contradicting the order-one separation of the endpoint. Carrying the constants through this argument yields the explicit statement $T > 4(N+1)[1 - a(\log q/q)^{2/3}]$ for every $a \geq 2$ and every $N \geq N_0(a)$, i.e. a coefficient approaching 4. The two layers combine into a single bound valid at every order (Corollary 2.4), and the asymptotic coefficient $\liminf_{N \rightarrow \infty} T_{\min}(N, \phi)/N \geq 4$ is then immediate.

The proof is also an explicit formalization target. The control model, admissibility, achievable costs and the scalar carrier, together with both main theorems – the elementary bound and the sharpened bound with its explicit threshold – are contained in a public Lean 4 + Mathlib development [21]. The present paper follows that formal structure closely, and Section 11 records statement by statement what is machine-checked and what is not.

2 Control model and exact mathematical statement

2.1 Elementary rotations and composite sequences

Let

$$X(\alpha) := \exp\left(-i\frac{\alpha}{2}\sigma_x\right), \quad Z(\theta) := \exp\left(-i\frac{\theta}{2}\sigma_z\right), \quad (1)$$

where σ_x and σ_z are the Pauli matrices. Throughout, $\|\cdot\|$ is the operator norm on matrices (the largest singular value), which equals one on unitary matrices, and all numerics reported in Section 8 use the same norm. The X rotations are assumed exact, instantaneous, and free. The Z evolution is the only costly operation, and in the presence of a fixed unknown fractional error ε its actual propagator is

$$Z^\varepsilon(\theta) = \exp\left[-i\frac{(1+\varepsilon)\theta}{2}\sigma_z\right]. \quad (2)$$

For $L \geq 0$, choose real angles α_j and nonnegative Z angles θ_j . Following the recursive order used in the Lean development, define

$$U_\lambda = X(\alpha_1)Z(\lambda\theta_1)X(\alpha_2)Z(\lambda\theta_2)\cdots X(\alpha_L)Z(\lambda\theta_L), \quad (3)$$

with $U_\lambda = I$ for $L = 0$. Relabelling the indices gives the equivalent convention in which $X(\alpha_L)Z(\lambda\theta_L)\cdots X(\alpha_1)Z(\lambda\theta_1)$ is written from left to right. Nothing in the proof depends on this notational choice.

The parameter λ is an interpolation variable: the physical propagator at error ε is $U_{1+\varepsilon}$. The target is

$$U_1 = Z(\phi), \quad 0 < \phi \leq \pi. \quad (4)$$

The cost is the total nominal Z angle,

$$T := \sum_{j=1}^L \theta_j. \quad (5)$$

The normalization is chosen so that the nominal Z rotation rate is one; restoring a physical rate ω simply replaces T by T/ω in the time interpretation.

Definition 2.1 (Order- N robustness). A sequence is *admissible of order N for ϕ* if $\theta_j \geq 0$, $U_1 = Z(\phi)$, and

$$\left.\frac{d^k U_\lambda}{d\lambda^k}\right|_{\lambda=1} = 0, \quad k = 1, \dots, N. \quad (6)$$

This is exactly matrix-valued flatness at the nominal error point.

Let $\text{costs}(N, \phi)$ denote the set of costs of all admissible sequences and define mathematically

$$T_{\min}(N, \phi) := \inf \text{costs}(N, \phi), \quad c_*(\phi) := \liminf_{N \rightarrow \infty} \frac{T_{\min}(N, \phi)}{N}. \quad (7)$$

The theorem below is understood with the usual extended-real convention: if an order has no admissible sequence, its minimum cost is $+\infty$. Section 11 records the exact real-valued Lean statement, where nonemptiness and an eventual boundedness hypothesis occur because of the definitions of `sInf` and `real Filter.liminf` in Mathlib.

Theorem 2.2 (Elementary bound, valid at every order). *For every $0 < \phi \leq \pi$ and every order- N admissible sequence,*

$$T \geq 2 \left[2(2N+2)! \sin^2\left(\frac{\phi}{4}\right) \right]^{1/(2N+2)}. \quad (8)$$

In particular, at $\phi = \pi$, $T \geq 2[(2N+2)!]^{1/(2N+2)}$, which exceeds $4(N+1)/e$ at every order; the limiting coefficient of N is $4/e \approx 1.4715$.

N	5	10	12	20	100	5010	10^6
elementary, $2[(2N+2)!]^{1/(2N+2)}$	10.58	18.11	21.10	33.02	151.3	7377.8	$1.47 \cdot 10^6$
sharpened, $a = 2.3$	4.7	16.6	22.1	45.5	321.9	$1.96 \cdot 10^4$	$4.00 \cdot 10^6$
$4(N+1)/e$	8.83	16.19	19.13	30.90	148.6	7373.8	$1.47 \cdot 10^6$

Table 1: The two lower bounds of Theorems 2.2 and 2.3, evaluated at sample orders N ; the sharpened entry is switched off below $N_0(a) = 5010$, so the small- N rows are shown only to locate the crossover at $N \approx 12$. The elementary bound already exceeds the closed form $4(N+1)/e$ at every order.

Theorem 2.3 (Sharpened bound, coefficient 4; main result). *Write $q = 2N + 2$. For every $a \geq 2$ there is an explicit threshold $N_0(a)$ such that every order- N admissible sequence with $N \geq N_0(a)$ and target $\phi = \pi$ satisfies*

$$T > 4(N+1) \left[1 - a \left(\frac{\log q}{q} \right)^{2/3} \right]. \quad (9)$$

Concretely $N_0(2.3) = 5010$ and $N_0(2.2) = 46000$; the exact definition of $N_0(a)$, together with the constants that enter it, is recorded in Section 11.

Corollary 2.4 (A single bound at every order). *Fix $a \geq 2$ and let $N_0(a)$ be as in Theorem 2.3. For every $N \geq 1$, every order- N admissible sequence with $\phi = \pi$ satisfies*

$$T \geq \max \left\{ 2[(2N+2)!]^{1/(2N+2)}, \mathbf{1}_{N \geq N_0(a)} \cdot 4(N+1) \left[1 - a \left(\frac{\log q}{q} \right)^{2/3} \right] \right\}. \quad (10)$$

Consequently $c_*(\pi) \geq 4$; more generally, whenever admissible sequences exist at every order and $0 < \phi \leq \pi$, the same whole-band argument gives $c_*(\phi) \geq 4$.

The two layers are compared at sample orders in Table 1. The sharpened term overtakes the elementary one near $N = 12$, and their ratio tends to e , reflecting the two limiting coefficients $4/e$ and 4 . Neither bound is asymptotic in character: each is an explicit inequality at the stated order, and (10) combines them into one.

The elementary bound is proved in Section 4; it is the estimate that holds at every order and needs no approximation theory. The sharpened bound is proved in Sections 5–6: it is established for an arbitrary $\rho < 1$ by contradiction, and the explicit dependence of the constants on a in $\rho = 1 - a(\log q/q)^{2/3}$ is what turns the argument into the finite-order statement (9). The existence hypothesis takes no part in the analytic argument: with the extended-real convention $\inf \emptyset = +\infty$ of (7) the conclusion holds vacuously whenever some order admits no admissible sequence. It is stated here because Mathlib’s real `sInf` assigns 0 to the empty set, so the formal statement has to assume nonemptiness explicitly (Section 11).

3 The scalar error carrier

Set

$$\tau := \frac{T}{2}, \quad q := 2N + 2, \quad (11)$$

and introduce the scalar carrier

$$h(\lambda) := 1 - \frac{1}{2} \operatorname{Re} \operatorname{Tr} \left(U_1^\dagger U_\lambda \right). \quad (12)$$

The choice of normalization is convenient because h simultaneously measures endpoint separation, exposes the doubled flatness order, and admits a finite-band exponential representation.

3.1 Finite exponential bandwidth

Because

$$Z(\lambda\theta) = e^{-i\lambda\theta/2} \frac{I + \sigma_z}{2} + e^{+i\lambda\theta/2} \frac{I - \sigma_z}{2},$$

expanding the product in (3) shows that every matrix element of U_λ is a finite sum of exponentials $e^{i\mu\lambda}$ with

$$\mu = \frac{1}{2} \sum_{j=1}^L s_j \theta_j, \quad s_j \in \{-1, +1\}. \quad (13)$$

Consequently,

$$h(\lambda) = \sum_m \gamma_m e^{i\mu_m \lambda}, \quad |\mu_m| \leq \tau. \quad (14)$$

Thus $\tau = T/2$ is a genuine frequency-bandwidth bound: no component of h oscillates faster than τ .

3.2 Bounds on the carrier

The relative propagator $V_\lambda := U_1^\dagger U_\lambda$ is unitary. Therefore $|\text{Tr } V_\lambda / 2| \leq 1$, and hence

$$0 \leq h(\lambda) \leq 2. \quad (15)$$

At the nominal point $\lambda = 1$, $V_1 = I$. Write locally

$$V_\lambda = a(\lambda)I + i\mathbf{b}(\lambda) \cdot \boldsymbol{\sigma}, \quad a^2 + |\mathbf{b}|^2 = 1,$$

with $a(1) = 1$ and $\mathbf{b}(1) = 0$. Matrix flatness of U_λ through order N implies flatness of V_λ through order N , so $\mathbf{b}$ has a zero of order at least $N + 1$. Since

$$h = 1 - a = \frac{1 - a^2}{1 + a} = \frac{|\mathbf{b}|^2}{1 + a}$$

near $\lambda = 1$, the zero order doubles.

Proposition 3.1 (Carrier flatness). *For an order- N admissible sequence,*

$$h^{(k)}(1) = 0, \quad k = 0, 1, \dots, q - 1, \quad q = 2N + 2. \quad (16)$$

In particular, the scalar flatness is a consequence of the matrix flatness and is not an additional assumption.

3.3 The endpoint at $\lambda = 0$

At $\lambda = 0$, all Z factors become the identity, and therefore

$$U_0 = X(A), \quad A := \sum_{j=1}^L \alpha_j. \quad (17)$$

The target endpoint is $U_1 = Z(\phi)$. A direct $SU(2)$ calculation gives

$$\text{Tr} \left[Z(\phi)^\dagger X(A) \right] = 2 \cos \frac{\phi}{2} \cos \frac{A}{2}. \quad (18)$$

Hence

$$\boxed{h(0) = 1 - \cos \frac{\phi}{2} \cos \frac{A}{2}}. \quad (19)$$

For $0 < \phi \leq \pi$, $\cos(\phi/2) \geq 0$, so

$$h(0) \geq 1 - \cos \frac{\phi}{2} = 2 \sin^2 \frac{\phi}{4} > 0. \quad (20)$$

The distinction between (19) and (20) is important: the equality $h(0) = 1 - \cos(\phi/2)$ is not generally true because the free X rotations leave the arbitrary factor $\cos(A/2)$.

There is also a useful geometric interpretation. For $U, V \in SU(2)$,

$$\|U - V\|^2 = 2 - \text{Tr}(U^\dagger V), \quad (21)$$

so, in the present case,

$$h(0) = \frac{1}{2} \|U_1 - U_0\|^2. \quad (22)$$

Thus the lower bound (20) says that the target is uniformly separated from the set of endpoint propagators obtainable by turning off the costly Z evolution.

4 Two elementary bounds: $2/e$ and $4/e$

The coefficient 4 of Theorem 2.3 is obtained from a substantially more elaborate whole-band argument. It is useful first to see that linear growth itself follows from very short calculations. The second of the two estimates below is Theorem 2.2, the bound that holds at every order, and it is machine-checked in the formal development (`elementary_bound`, `elementary_bound_pi`, Section 11); the weaker $2/e$ propagator estimate of the first subsection is paper-level and is superseded by it.

4.1 Direct propagator estimate: $2/e$

Differentiating a single Z factor gives

$$\left\| \frac{d^m}{d\lambda^m} Z(\lambda\theta_j) \right\| = \left(\frac{\theta_j}{2} \right)^m.$$

Applying the product rule to the whole sequence and using the unitarity of all undifferentiated factors gives the multinomial estimate

$$\|U_\lambda^{(k)}\| \leq \sum_{k_1 + \dots + k_L = k} \frac{k!}{k_1! \dots k_L!} \prod_j \left(\frac{\theta_j}{2} \right)^{k_j} = \tau^k. \quad (23)$$

Taylor's theorem about $\lambda = 1$, using the first N vanishing derivatives, gives

$$\|U_0 - U_1\| \leq \frac{\tau^{N+1}}{(N+1)!}. \quad (24)$$

On the other hand, $U_0 = X(A)$ and $U_1 = Z(\phi)$, and the exact endpoint geometry gives

$$\|X(A) - Z(\phi)\| \geq 2 \sin \frac{\phi}{4}. \quad (25)$$

Therefore every admissible sequence satisfies

$$T \geq 2 \left[2(N+1)! \sin\left(\frac{\phi}{4}\right) \right]^{1/(N+1)}. \quad (26)$$

Using Stirling's formula, $(N+1)!^{1/(N+1)} = (N+1)e^{-1}(1+o(1))$, and the fixed factor $[2\sin(\phi/4)]^{1/(N+1)}$ tends to one. Hence

$$c_*(\phi) \geq \frac{2}{e}. \quad (27)$$

This is already enough to rule out any sublinear robustification. In particular, the conclusion that arbitrarily high-order cancellation must cost at least $\Omega(N)$ does not depend on the technical machinery of the main proof.

4.2 Using the doubled flatness: $4/e$

The improvement is immediate once the scalar carrier is introduced. From (23) and $|\operatorname{Tr} M| \leq 2\|M\|$, one obtains

$$|h^{(k)}(\lambda)| \leq \tau^k. \quad (28)$$

By Proposition 3.1, the first $q-1$ derivatives of h vanish at 1, where $q = 2N+2$. Taylor's theorem therefore yields

$$|h(0)| \leq \frac{\tau^q}{q!}. \quad (29)$$

Combining with (20) gives

$$T \geq 2 \left[2q! \sin^2\left(\frac{\phi}{4}\right) \right]^{1/q}, \quad q = 2N+2. \quad (30)$$

Since $(q!)^{1/q} \sim q/e$, more precisely $(q!)^{1/q} = (q/e)(1+o(1))$,

$$c_*(\phi) \geq \frac{4}{e} \approx 1.4715. \quad (31)$$

The factor of two in the asymptotic coefficient comes entirely from the doubled flatness order $q = 2(N+1)$. No new physical assumption is introduced. The only information not yet exploited is that h is a band-limited finite exponential sum. The main theorem obtains a further factor e by using that whole band rather than one Taylor remainder point.

5 The whole-band argument

5.1 Vanishing derivatives become vanishing moments

Let

$$h(\lambda) = \sum_m \gamma_m e^{i\mu_m \lambda}, \quad |\mu_m| \leq \tau.$$

Differentiation gives

$$h^{(k)}(1) = i^k \sum_m \gamma_m \mu_m^k e^{i\mu_m}.$$

Hence Proposition 3.1 is equivalent to

$$\boxed{\sum_m \gamma_m \mu_m^k e^{i\mu_m} = 0, \quad k = 0, 1, \dots, q-1.} \quad (32)$$

These are the moment constraints that drive the approximation argument.

If P is any polynomial of degree strictly smaller than q , then the moments imply

$$\sum_m \gamma_m P(\mu_m) e^{i\mu_m} = 0. \quad (33)$$

Since $h(0) = \sum_m \gamma_m$, we can therefore write

$$\boxed{h(0) = \sum_m \gamma_m \left[1 - P(\mu_m) e^{i\mu_m} \right].} \quad (34)$$

Moreover,

$$\left| 1 - P(\mu) e^{i\mu} \right| = \left| e^{-i\mu} - P(\mu) \right|. \quad (35)$$

Thus the endpoint value can be made small whenever $e^{-i\mu}$ can be approximated on the entire band $[-\tau, \tau]$ by a polynomial of degree $< q$ with sufficiently small error.

There is one technical issue: the identity (34) is a discrete sum, whereas a useful uniform approximation is easier to implement through Fourier inversion. The next subsection gives the required smoothing step.

5.2 Smoothing by a compactly supported Fourier multiplier

Let P be a degree- $(q-1)$ approximation and define on the band

$$\Psi(\mu) := 1 - P(\mu) e^{i\mu}. \quad (36)$$

Choose a C^2 cutoff χ satisfying

$$\chi(\mu) = 1 \quad (|\mu| \leq \tau), \quad \chi(\mu) = 0 \quad (|\mu| \geq \tau + B), \quad (37)$$

with a fixed absolute cutoff constant K such that

$$|\chi'(\mu)| \leq \frac{K}{B}, \quad |\chi''(\mu)| \leq \frac{K}{B^2}. \quad (38)$$

Set

$$\widehat{\kappa}(\mu) := \Psi(\mu) \chi(\mu), \quad \kappa(t) := \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\kappa}(\mu) e^{-i\mu t} d\mu. \quad (39)$$

Because $\widehat{\kappa}$ is compactly supported and C^2 , two integrations by parts give

$$|\kappa(t)| \leq \min\{C_1, C_2/t^2\}, \quad (40)$$

where

$$C_1 := \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{\kappa}(\mu)| d\mu, \quad C_2 := \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{\kappa}''(\mu)| d\mu. \quad (41)$$

Splitting the t integral at $|t| = \sqrt{C_2/C_1}$ yields the elementary estimate

$$\|\kappa\|_{L^1} \leq 4\sqrt{C_1 C_2}. \quad (42)$$

On the frequency band, $\chi = 1$, so (14) gives

$$\int_{\mathbb{R}} h(t) \kappa(t) dt = \sum_m \gamma_m \hat{\kappa}(\mu_m) \quad (43)$$

$$= \sum_m \gamma_m \left[1 - P(\mu_m) e^{i\mu_m} \right] = h(0). \quad (44)$$

Together with $\|h\|_{\infty} \leq 2$, this gives

$$\boxed{|h(0)| \leq 8\sqrt{C_1 C_2}}. \quad (45)$$

Thus the problem has been reduced to controlling the product $C_1 C_2$.

5.3 Polynomial approximation on the frequency band

The natural polynomial comes from the Fourier representation of $e^{-i\tau \cos u}$. Define

$$c_n(\tau) := \frac{1}{2\pi} \int_0^{2\pi} e^{-inu} e^{-i\tau \cos u} du. \quad (46)$$

Then

$$e^{-i\tau \cos u} = c_0(\tau) + 2 \sum_{n \geq 1} c_n(\tau) T_n(\cos u), \quad (47)$$

so, on $|\mu| \leq \tau$,

$$e^{-i\mu} = c_0(\tau) + 2 \sum_{n \geq 1} c_n(\tau) T_n\left(\frac{\mu}{\tau}\right). \quad (48)$$

Equivalently $c_n(\tau) = i^{-n} J_n(\tau)$, so the degree at which the truncation must stop is the Bessel turning point $n \approx \tau$; the formal development does not introduce Bessel functions and works directly with the Fourier coefficients.

For any $\delta > 0$, shift the contour from u to $u - i\delta$; the integrand is entire and 2π -periodic, so the integral is unchanged. Since

$$\cos(u - i\delta) = \cos u \cosh \delta + i \sin u \sinh \delta,$$

the two factors of the shifted integrand obey $|e^{-in(u-i\delta)}| = e^{-n\delta}$ and $|e^{-i\tau \cos(u-i\delta)}| = e^{\tau \sin u \sinh \delta} \leq e^{\tau \sinh \delta}$, so for $n > 0$

$$|c_n(\tau)| \leq \exp(\tau \sinh \delta - n\delta). \quad (49)$$

Truncate the series after degree $q-1$ and denote the resulting polynomial by P_{q-1} . A geometric tail estimate gives an approximation error bounded by

$$\eta := \sup_{|\mu| \leq \tau} |e^{-i\mu} - P_{q-1}(\mu)| \leq \left(1 + \frac{4}{1 - e^{-\delta}}\right) e^{\delta} \exp(\tau \sinh \delta - \delta q). \quad (50)$$

The precise form of the harmless prefactor is chosen here to match the estimate used in the formal development.¹

Suppose now that

$$\tau \leq \rho q, \quad 0 < \rho < 1. \quad (51)$$

¹The symbol η always denotes this band-approximation error. The Lean source calls the same quantity ε ; that name is not used for it here, since ε already denotes the fractional error (2).

Choose $\delta' > 0$ so small that

$$\eta_0 := \delta' - \rho \sinh \delta' > 0. \quad (52)$$

Such a choice exists because $\sinh \delta' = \delta' + O((\delta')^3)$ and $\rho < 1$. Then

$$\eta \leq D_0 e^{-\eta_0 q}, \quad D_0 := \left(1 + \frac{4}{1 - e^{-\delta'}}\right) e^{\delta'}. \quad (53)$$

The error therefore decays exponentially in the polynomial degree as long as the frequency bandwidth stays below the critical ratio $\tau/q = 1$.

5.4 The τ -scaled collar

The polynomial approximation is required not only on the band itself. The Fourier cutoff extends the multiplier to the collar $\tau < |\mu| \leq \tau + B$, so derivatives of P must remain controlled there. A collar width that is fixed independently of τ is not uniform when $\tau \rightarrow 0$. The formalized solution is to choose the width relative to the band:

$$B := \frac{\eta\tau}{q^2}. \quad (54)$$

Thus

$$\frac{B}{\tau} = \frac{\eta}{q^2}, \quad (55)$$

which is independent of the absolute scale of τ . Both B and the error η are dimensionless, since angles are measured in units of the nominal Z rate.

The Chebyshev polynomials satisfy the differential equation

$$(1 - u^2)T_n''(u) = uT_n'(u) - n^2T_n(u). \quad (56)$$

Using this equation together with the usual collar growth bound gives, for $|u| \leq 1 + \beta$ with $0 \leq \beta \leq 1/2$,

$$|T_n''(u)| \leq 32n^4 e^{2n\sqrt{\beta}}. \quad (57)$$

With the choice $\beta = B/\tau = \eta/q^2$, and the coefficient decay (49) on the tail $n \geq q$ of the Chebyshev series, this controls P , P' , and P'' uniformly on the whole collar. Two side conditions enter and both hold for all sufficiently large q , because η decays exponentially in q while $\beta = \eta/q^2$: the collar bound requires $\beta \leq 1/2$, and the collar growth $e^{2n\sqrt{\beta}}$ must be beaten by the decay, that is $\delta > 2\sqrt{\beta}$.

The powers of q come from the head of the series, $n < q$, where one uses only $|c_n(\tau)| \leq 1$ — immediate from (46), whose integrand has modulus one — and not the exponentially large prefactor of (49); this is what keeps the collar estimates polynomial in q rather than exponential. The product rule applied to $\hat{\kappa} = \Psi\chi$ then gives bounds for $\hat{\kappa}$ and $\hat{\kappa}''$: on the band $|\hat{\kappa}| \leq \eta$, and on the collar the cutoff contributes the factors K/B and K/B^2 of (38), while both integrals are taken over the support, of length at most $2(\tau + B)$. Appendix E records this assembly. The resulting calculation is technical but elementary; its important feature is the scaling in q and η , not the absolute numerical constant. The formal Lean development proves the product estimate

$$C_1 C_2 \leq 3000(1 + K) \eta q^8, \quad (58)$$

where K is the fixed cutoff constant. Combining with (53) gives

$$C_1 C_2 \leq 3000(1 + K) \left(1 + \frac{4}{1 - e^{-\delta'}}\right) e^{\delta'} q^8 e^{-\eta_0 q}. \quad (59)$$

This is the quantitative heart of the formal proof: all dependence on τ has disappeared except for the hypothesis $\tau \leq \rho q$.

6 Proof of the main theorem

We now assemble the preceding ingredients exactly in the order implemented by the Lean formalization.

Fix $0 < \rho < 1$. Assume, for contradiction, that there are arbitrarily large orders N with an admissible sequence satisfying

$$\tau = \frac{T}{2} \leq \rho(2N + 2) = \rho q. \quad (60)$$

Choose $\delta' > 0$ such that $\eta_0 = \delta' - \rho \sinh \delta' > 0$. The polynomial approximation above then satisfies (53), and the τ -scaled collar estimate gives (59). Hence (45) yields

$$|h(0)| \leq 8 \sqrt{3000(1 + K) \left(1 + \frac{4}{1 - e^{-\delta'}}\right) e^{\delta'} q^8 e^{-\eta_0 q}} \xrightarrow{q \rightarrow \infty} 0. \quad (61)$$

In particular, for all sufficiently large N ,

$$h(0) < 1 - \cos \frac{\phi}{2}. \quad (62)$$

But the exact endpoint identity (19) implies the opposite inequality

$$h(0) \geq 1 - \cos \frac{\phi}{2}, \quad (63)$$

which is a contradiction.

We have therefore proved the key statement

$$\boxed{\tau \leq \rho(2N + 2) \text{ is impossible for all sufficiently large } N.} \quad (64)$$

Equivalently, every sufficiently high-order admissible sequence satisfies

$$T > 2\rho(2N + 2) = 4\rho(N + 1).$$

Taking the infimum over admissible sequences preserves the lower bound, and letting $\rho \uparrow 1$ gives

$$\liminf_{N \rightarrow \infty} \frac{T_{\min}(N, \phi)}{N} \geq 4.$$

This proves the mechanism behind Theorem 2.3; the explicit threshold follows by carrying the constants through the same argument (Section 11).

6.1 From the mechanism to an explicit threshold

The argument above is stated for an arbitrary $0 < \rho < 1$, and the polynomial factor q^8 in (61) is what makes the resulting threshold implicit. For the finite-order statement (9) the same chain is used with sharper constants; only two of them matter.

First, the collar bounds on the band error can be made *linear* in the approximation error ε instead of absolute. Writing $k = 2N + 1$ for the truncation degree and $\Psi_A = e^{-i\mu} - P(\mu)$, the truncated Chebyshev series satisfies, on the whole collar $|\mu| \leq \tau + B$,

$$\sup |\Psi'_A| \leq \frac{c_1 k^2}{\tau} \varepsilon, \quad \sup |\Psi''_A| \leq \frac{c_2 k^4}{\tau^2} \varepsilon, \quad (65)$$

with explicit constants $c_1 = 7$, $c_2 = 384$. The exponents k^2 and k^4 are the correct scalings for this route – the tail of the series contributes an n^2 -weighted sum, so a k -scaling is false with any constant independent of k , while the collar growth $e^{k \cdot 2\sqrt{B/\tau}}$ is bounded by an absolute factor. Inserting (65) into the support-aware bounds for C_1 and C_2 of Section 5, with the cutoff scale $B = \sqrt{\varepsilon}$ and $k \leq q$, gives

$$C_1 C_2 \leq C_{\sharp} q^{\beta} \varepsilon^2, \quad \beta \leq 4, \quad (66)$$

with $C_{\sharp}$ explicit. Compared with (58) the power of q has dropped from 8 to at most 4, and the error enters quadratically.

Second, the approximation error is exponential. With $\delta' = \operatorname{arccosh}(1/\rho)$ and $\mu(\rho) = \delta' - \sqrt{1 - \rho^2}$,

$$\varepsilon \leq D_0(\rho) e^{-q\mu(\rho)}, \quad D_0(\rho) = \left(1 + \frac{4}{1 - e^{-\delta'}}\right) e^{\delta'}. \quad (67)$$

The contradiction of (45) and (63) therefore holds as soon as $D_0(\rho) e^{-q\mu(\rho)}$ is smaller than an explicit q -dependent threshold, and that is an inequality in q alone. For the profile

$$\rho(q) = 1 - a \left(\frac{\log q}{q} \right)^{2/3}, \quad a \geq 2, \quad (68)$$

and using $\mu(1 - \varepsilon_0) \geq \frac{47}{50} \varepsilon_0^{3/2}$ and $D_0(\rho) \leq 13(1 - \rho)^{-1/2}$ together with the explicit constant $K_0 = 21.1$ of the smoothing multiplier, the criterion reduces to $q \geq N_0(a)$ with $N_0(a)$ explicit; the two instances quoted in Theorem 2.3 are $N_0(2.3) = 5010$ and $N_0(2.2) = 46000$. Section 11 records the definitions, the constants, and the machine-checked status of every step.

Remark 6.1 (Where the coefficient 4 comes from). The constant is transparent once the proof is compressed to one line:

$$q = 2N + 2, \quad \tau = \frac{T}{2}, \quad \tau < q \implies h(0) \longrightarrow 0.$$

The largest bandwidth ratio for which the approximation argument gives exponential suppression is therefore $\tau/q < 1$. Restoring $T = 2\tau$ and $q \sim 2N$ gives $T/N \geq 2 \times 2 = 4$. The number 4 is thus the threshold delivered by this proof mechanism. The theorem does not claim that the optimal control problem itself has asymptotic coefficient exactly 4.

7 Target angle and scope

The formal theorem is stated for $0 < \phi \leq \pi$, exactly as in the Lean statement. The endpoint lower bound (20) is the only place where the target angle enters the proof, and it is strictly positive throughout this interval.

At $\phi = \pi$ the endpoint separation is maximal:

$$h(0) = 1 \quad \text{for every } A, \quad (69)$$

because $\cos(\pi/2) = 0$. Thus $Z(\pi)$ is the target that is furthest, in the relevant endpoint metric, from the set of pure X rotations.

At the other trivial endpoint, $\phi = 0$, the target is I and zero-cost control already realizes it exactly. Likewise $Z(2\pi) = -I$ differs only by a global phase and can be generated by the free rotation $X(2\pi) = -I$. Hence no positive lower bound can extend to the endpoints where the target becomes trivial up to global phase.

The present paper does not need an extension beyond $0 < \phi \leq \pi$. Such extensions can be treated by additional unitary symmetries, but they are not part of the machine-checked theorem and are therefore not used in the main claim.

8 Numerical upper bounds for the hardest target

Corollary 2.4 gives $c_*(\phi) \geq 4$ and leaves the upper end of the window open. The gap is quantitative and worth measuring, and the target to measure it on is $\phi = \pi$: its endpoint separation (69) is the largest, so it is the target at which the endpoint obstruction (20) is largest, and it is the target realized in the Rydberg protocol of the unpublished manuscript [20], where each segment of free evolution is designed to accumulate the phase π .

The section is one-sided by construction and should be read that way. It reports the costs of sequences that were found by a local search and then verified to be admissible. Every number is therefore an achievable cost, hence an upper bound for $T_{\min}(N, \pi)$, and no number here is certified minimal or extrapolated to large N ; the precise status of the data is collected at the end of the section, before the discussion.

For a fixed number L of Z segments, write the sequence in the form (3),

$$U_\lambda = X(\alpha_1)Z(\lambda\theta_1)X(\alpha_2)Z(\lambda\theta_2)\cdots X(\alpha_L)Z(\lambda\theta_L),$$

and consider the finite-dimensional problem

$$T_{\min}(N, \pi) \leq \min \left\{ \sum_{j=1}^L \theta_j \right\}, \quad \theta_j \geq 0, \quad U_1 = Z(\pi), \quad \left. \frac{d^k U_\lambda}{d\lambda^k} \right|_{\lambda=1} = 0, \quad k \leq N. \quad (70)$$

The minimum is taken over every segment number L and every admissible angle list, so any admissible angle list here satisfies Definition 2.1 and its cost is an upper bound for $T_{\min}(N, \pi)$ — not for the asymptotic coefficient $c_*(\pi)$, which no single order can bound. Nothing in (70) restricts the amplitudes to be equal, and that restriction is the only reason the equiangular family of [3] costs $T = (2N + 1)\pi$.

How the numbers were obtained and checked. The minimizations used a truncated-Taylor propagator in the error variable, in which the derivative conditions (6) hold exactly within the truncated model, and a time-budget parametrization that turns (70) at fixed T into a feasibility problem, pushed down by an adaptive march and warm-started by continuation in the segment number and in the order. Every solution reported in Table 2 was then re-verified from scratch, with the three conditions kept apart:

- (i) *Nominal identity.* The nominal propagator U_1 is rebuilt with an independent `scipy.linalg.expm` product, one factor at a time and with no Taylor algebra, and compared with $Z(\pi)$ up to the global sign $-I = X(2\pi)$. That sign is harmless here: a sequence realizing $-Z(\pi)$ realizes $Z(\pi)$ after the free outer rotation $X(2\pi)$, with the same cost and the same derivatives. The mismatch $\|U_1 \mp Z(\pi)\|$ is at most 9×10^{-10} for every order $N \leq 12$, which is the accuracy to which the angles themselves are determined; it is the tolerance of the search, not the round-off of double precision.
- (ii) *Flatness.* Write $u_k = d^k U_\lambda / d\lambda^k$ at $\lambda = 1$. Since the a priori bound (23) is $\|u_k\| \leq \tau^k$, the meaningful residual is the dimensionless ratio $\rho_N := \max_{1 \leq k \leq N} \|u_k\| / \tau^k$; it is at most 1.5×10^{-10} in every row of Table 2. The cancellation is therefore complete to the same ten digits as the nominal identity, whereas for a generic sequence of the same cost the norms would be of order τ^k : at $N = 12$ the largest one is 1.1×10^3 , against $\tau^{12} \approx 2 \times 10^{18}$. The coefficients $[\varepsilon^k]U$ are evaluated twice, by a truncated convolution in double precision and by an exact matrix series at 50 decimal digits; the two agree on ρ_N to better than 1%, the finite precision of the first being the limiting factor, which is why the tabulated values come from the 50-digit run.

Table 2: One admissible sequence per order at $\phi = \pi$, for $N \leq 12$. Each row is the cost $T = \sum_j \theta_j$ of the sequence printed in Appendix G, which uses L segments, some of them of zero length, so $T_{\min}(N, \pi) \leq T$; no row is claimed to be optimal or to be a certified minimum. The ratio column compares T with the equiangular cost $(2N + 1)\pi$ of [3]. The last three columns are the independent re-verification described in the text: the nominal mismatch, the dimensionless flatness residual $\rho_N = \max_{k \leq N} \|u_k\|/\tau^k$ of (23) with $\tau = T/2$, and the last three local logarithmic slopes of $\delta(\varepsilon)$, to be read against $N + 1$.

N	L	T	T/N	$T/(2N+1)\pi$	$\ U_1 \mp Z(\pi)\ $	ρ_N	slopes of δ
1	17	8.1505	8.150	0.865	$2.0 \cdot 10^{-10}$	$4.0 \cdot 10^{-11}$	1.98, 1.99, 2.00
2	21	13.3466	6.673	0.850	$2.5 \cdot 10^{-10}$	$1.0 \cdot 10^{-10}$	2.96, 2.98, 3.00
3	25	18.5505	6.183	0.844	$2.5 \cdot 10^{-12}$	$2.7 \cdot 10^{-12}$	3.94, 3.97, 3.99
4	31	23.7878	5.947	0.841	$1.6 \cdot 10^{-11}$	$4.5 \cdot 10^{-12}$	4.91, 4.96, 4.99
5	38	29.0593	5.812	0.841	$3.3 \cdot 10^{-10}$	$4.0 \cdot 10^{-11}$	5.89, 5.95, 5.99
6	48	34.2512	5.709	0.839	$3.3 \cdot 10^{-12}$	$1.4 \cdot 10^{-12}$	6.87, 6.94, 6.98
7	55	39.7030	5.672	0.843	$1.2 \cdot 10^{-10}$	$1.3 \cdot 10^{-11}$	7.84, 7.92, 7.93
8	58	44.9604	5.620	0.842	$4.3 \cdot 10^{-10}$	$3.4 \cdot 10^{-11}$	8.82, 8.91, 7.99
9	63	52.2657	5.807	0.876	$1.9 \cdot 10^{-11}$	$2.8 \cdot 10^{-12}$	9.63, 9.74, 4.43
10	70	56.9380	5.694	0.863	$7.0 \cdot 10^{-11}$	$1.2 \cdot 10^{-11}$	10.76, 10.58, 1.67
11	75	66.9068	6.082	0.926	$7.7 \cdot 10^{-10}$	$1.4 \cdot 10^{-10}$	11.66, 7.59, 0.70
12	78	67.6302	5.636	0.861	$8.7 \cdot 10^{-10}$	$7.8 \cdot 10^{-11}$	12.25, 12.51, 12.68

- (iii) *Order of robustness.* The deviation from the target, $\delta(\varepsilon) = \arccos(|\langle U_{1+\varepsilon}, Z(\pi) \rangle|/2)$ with $\langle U, V \rangle = \text{Tr}(U^\dagger V)$, is evaluated on a grid of errors above the round-off floor. Its local logarithmic slope rises towards $N + 1$ as ε decreases and then saturates once δ reaches that floor: the last three slopes are 1.98, 1.99, 2.00 at $N = 1$, 8.82, 8.91, 7.99 at $N = 8$, and 12.25, 12.51, 12.68 at $N = 12$. From $N = 8$ on the last entries of the column fall short of $N + 1$ for that reason, and no order is inferred from them. Where the slope is resolved, the cancellation is carried to order N and stops there, which is exactly what (6) asks for.

What the numbers say. Write $T_{\text{num}}(N, \pi)$ for the cost of the sequence found at order N ; by construction $T_{\min}(N, \pi) \leq T_{\text{num}}(N, \pi)$, and the two need not be equal. Over the range in which the search depth is uniform, $N \leq 8$, those upper bounds increase by almost exactly 5.2 per order,

$$T_{\text{num}}(N, \pi) \approx 5.26 N + 2.81,$$

with increments 5.20, 5.20, 5.24, 5.27, 5.19, 5.45, 5.26 and residuals below 0.119. The equiangular cost is beaten by a factor between 0.838 and 0.926 throughout. All the cost comparison gives at small order is $T/N < (2 + 1/N)\pi$, since $T < (2N + 1)\pi$: the coefficient T/N is above 2π at $N = 1, 2$ and stays below it from $N = 3$ on, reaching its smallest value 5.62 at $N = 8$. Two consequences are worth separating.

First, the gain over the equiangular cost may come from the amplitudes alone: that family fixes every amplitude at π for algebraic convenience, while (70) leaves them free, and every cost found is below $(2N + 1)\pi$. How much of the gain is available in this way is a question these numbers do not answer. Second, the found costs stay well above the proved floor 4; whether the true coefficient is closer to 4 or to the values found here is precisely what these upper bounds cannot decide. The last four rows illustrate the limits of the search rather than a property of the problem: they were seeded from the equiangular family, and their increment pattern is irregular (7.31 after $N = 8$, then 4.67, 9.97), which is a statement about the optimisation basin and not about $T_{\min}$. What the extended table does establish is that every cost listed is strictly below the equiangular cost $(2N + 1)\pi$, by a factor between 0.838 and 0.926, and that every ratio T/N is strictly above the proved floor 4.

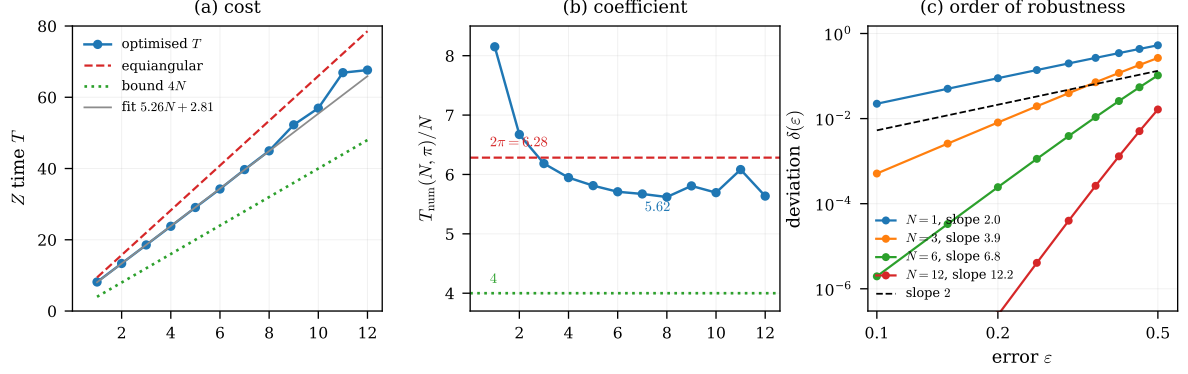


Figure 1: The same data as Table 2. (a) the optimised cost against the equiangular cost $(2N + 1)\pi$ and the lower bound $4N$; the straight line is the fit $T \approx 5.26N + 2.81$ to the branch $N \leq 8$, whose residuals are below 0.12. (b) the same cost divided by N , against the proved lower bound 4 and the equiangular coefficient $(2N + 1)\pi/N$; the dashed line is $2\pi \approx 6.28$, which the values at $N = 1, 2$ exceed. (c) the deviation $\delta(\varepsilon)$ on a log-log grid for $N = 1, 3, 6, 12$, with its local slope; the dashed line has slope 2, the smallest order appearing in the figure, and each curve is drawn only while δ stays above the round-off floor.

Status of the numbers, and what they do not claim. The optimisation is a local search: a damped least-squares march from warm starts, with no certificate of optimality and no search over segment numbers beyond continuation, and the residuals it drives to zero are feasibility conditions, not a minimality condition. What Section 8 verifies is *only* that each printed sequence is admissible: it realizes $Z(\pi)$ and it cancels the first N derivatives. Every entry of Table 2 is therefore a genuine cost of a genuine sequence, that is, an upper bound for $T_{\min}(N, \pi)$, and nothing more. In particular:

- The values are *not* claimed to be optimal. A cheaper sequence of the same order may exist, and $T_{\min}(N, \pi)$ may be strictly smaller than every number listed; no lower bound at any finite order is obtained here.
- No asymptotic claim is made. The fit $T \approx 5.26N + 2.81$ describes eight upper bounds with residuals below 0.119; it is not an estimate of $c_*(\pi)$, which these numbers cannot bound from either side. Because the quality of the search is itself N -dependent—the rows $N \geq 9$ come from a different seeding and are visibly looser—the increments of Table 2 cannot be extrapolated in N .
- The segment numbers L are those found by the search, not the smallest possible ones, and the comparison with the equiangular family is made at equal order N only.

Nothing in this section is used in the proofs of Theorems 2.2 and 2.3. The one statement the data do establish is one-sided and independent of optimality: the equiangular cost $(2N + 1)\pi$ is not optimal at any order considered, since every listed cost is strictly below it, by a factor between 0.838 and 0.927. Closing the window (78) therefore requires a construction in the free- X cost model that is genuinely different from the equiangular family—or a proof that no such construction reaches 4.

Interactive data. The sequences and their verification can be inspected in a browser at the first resource of Table 3: the page animates the Bloch-sphere trajectory of every sequence of Table 2, prints the full angle lists in units of π , and recomputes the nominal mismatch, the deviation $\delta(\varepsilon)$ and its slope in the browser from the same definitions as (1)–(6). The numbers

Table 3: Public resources for this paper. The Lean development is self-contained: a fresh clone reproduces its statements with the three commands of its `README`. The independent re-verification is likewise reproducible from the last entry, and Appendix G carries the same sequences at the precision used there. All addresses were accessed on 8 October 2026.

Resource	Content
https://show.dytchem.cn/bloch/	Bloch-sphere animation and the full angle lists of every sequence of Table 2; browser-side re-computation of the nominal mismatch, the deviation $\delta(\varepsilon)$ and its slope, with the equiangular family shown side by side. [22]
https://show.dytchem.cn/lean/	Verification status of the formal development: the exact Lean statements of the finite-order bounds, the source scan for <code>sorry</code> and <code>axiom</code> , and the raw <code>lake build</code> and <code>#print axioms</code> output for the headline theorems (<code>elementary_bound_pi</code> , <code>endgame_bound_2_3_num</code> , <code>all_N_bound_end</code> , <code>c_ge_four</code>). [21]
GitHub repository	Lean 4.34.1 with Mathlib v4.34.1, pinned by <code>lean-toolchain</code> and <code>lake-manifest.json</code> ; <code>VERIFICATION.md</code> records two from-scratch verifications on independent machines. <code>git clone</code> , <code>lake exe cache get</code> , <code>lake build</code> : 3739 jobs, with no <code>sorry</code> or <code>axiom</code> in the sources. [21]
Verification scripts and data	<code>verify_num.py</code> , which rebuilds every propagator with <code>scipy.linalg.expm</code> and re-evaluates the Taylor coefficients of Section 8 at 50 decimal digits; <code>make_outputs.py</code> , which regenerates Figures 1 and 2 and Appendix G from the verified data; the data themselves (<code>num_results.json</code> , <code>solutions_largeN.json</code>); and a <code>README</code> . [22]

it reports are those tabulated here.

9 Code, data, and formal development

Everything used in this paper is public: the proofs are a Lean development, the numerical sequences and their independent re-verification are a data set with a browser front end, and the figures and tables of Section 8 are regenerated from that data by short scripts. Table 3 collects the four entry points. Neither the proof nor the verification depends on the demo pages, which are presentation only.

The two runs recorded for the formal development start from the public clone alone, with no files, caches or toolchains copied between machines, and the axiom audit on the main statements reports only `propext`, `Classical.choice` and `Quot.sound`. Appendix G lists the sequences in the convention in which they were verified, rounded to 12 decimals of π ; the rounding is checked, not assumed: feeding those rounded lists back into (1)–(6) again gives a nominal mismatch below 9×10^{-10} and a flatness residual below 1.5×10^{-10} , which are the two accuracy statements of Table 2.

10 Relation to previous work

The earliest composite-pulse constructions showed that systematic pulse-length errors can be cancelled to arbitrarily high order [1], and [2] reviews the vocabulary of order- N robustness and of cost accounting in that setting. Low, Yoder, and Chuang subsequently proved that, in a standard primitive-pulse model, the number of primitive rotations required for arbitrary accuracy has optimal linear scaling in the robustness order, with constructions saturating that

scaling [3]. Their statement has the same shape as Theorem 2.3 — a linear lower bound in the robustness order together with a construction that saturates it — but the resource is the number of primitive rotations, in a model where those rotations are themselves part of the costly control sequence. It therefore does not determine the minimum Z -evolution time in the present free- X model, which is what Theorem 2.3 bounds.

The identification of order- N error cancellation with the vanishing of the first N moments of the error is standard in dynamical decoupling and in the filter-function description of open-loop control: it is the defining condition of Uhrig’s optimized sequences [5], and it was put on a general footing by the transfer-function analysis of [6]. The composite-pulse counterpart, a single-qubit sequence analysed order by order in its systematic pulse-length error, is [4]. Section 5 uses the same identity — vanishing derivatives of the propagator at the nominal point are vanishing moments of the carrier — but the carrier here is a function of a Z -evolution budget rather than of a pulse train, and it is the additional information that this carrier is band-limited, not the identity itself, that produces the constant 4.

Kukita, Kiya, and Kondo derived a lower bound on operation time for first-order pulse-length-error robust composite gates using a geometric argument [7]. The resource model there assigns time to the rotations that change the control axis. In the present problem those rotations are idealized as instantaneous and free, so a geometric lower bound on the full control path does not immediately become a lower bound on T .

Two recent works bound noise-robust control from the side of the control amplitude. Zeng and Deng prove a universal average-case time lower bound for first-order robustness against a single quasi-static coherent error under bounded control amplitude, together with a construction that is robust in time $4T$ plus a constant [8]; that overhead is fixed and N -independent, whereas the coefficient studied here is linear in the robustness order. Kosut, Lidar, and Rabitz prove a universal lower bound on the fidelity of any robustly controlled gate as a function of the gate duration times an aggregate uncertainty [9]. Both are statements about first-order or order-independent robustness, and neither is sharp in the way Theorem 2.3 is for the model of Section 2: the present proof uses the analytic dependence on a common scale parameter λ and remains valid for arbitrary order N .

Machine-checked quantum information is by now a developed subject — quantum programs and compilers in Coq and Isabelle/HOL [10], and quantum information theory in Lean [11] — but we are not aware of a formalized lower bound in quantum control. The development behind Theorem 2.3 is an attempt in that direction, for a model small enough that the whole proof, analytic estimates included, is checked by the kernel.

The connection to the supplied Rydberg QSP manuscript [20] — unpublished at the time of writing — is complementary. That work treats an unknown interaction phase as the input to a QSP-style composite protocol and uses fast control pulses to suppress systematic interaction variation. The present result identifies a resource limitation in the idealized single-axis core of that philosophy: even with perfect free transverse controls, high-order cancellation cannot be made sublinear in the amount of uncertain evolution. No quantitative claim in the Rydberg manuscript is used in the proof of Theorem 2.3.

11 Exact correspondence with the Lean formalization

The statements of Section 2 are the paper form of named Lean 4 declarations of the public development [21]; the model (the rotations, the evolution and `Admissible`), the cost $T = \text{cost } \theta$ and the carrier h of (12) are fixed in `RobustZ/Statement.lean`, and the scalar flatness (16) is derived from the matrix flatness rather than assumed. Write $q = 2N + 2$, $\tau = T/2$ and

$k = 2N + 1 = q - 1$; the sharpened layer uses $\rho_a(q) = 1 - a(\log q/q)^{2/3}$, the exponent $\delta' > 0$ with $\cosh \delta' = 1/\rho_a(q)$, the Chebyshev tail size $\varepsilon = \text{eps0f } \tau \delta' k$, and the collar half-width $B = \sqrt{\varepsilon}$. The endgame statements are stated for $\phi = \pi$, the case formalized.

11.1 The machine-checked statements

Theorem 11.1 (Elementary bound, all orders; `elementary_bound`, `elementary_bound_pi`). *For every $0 < \phi \leq \pi$ and every order- N admissible sequence,*

$$T \geq 2(2q! \sin^2(\phi/4))^{1/q}, \quad \text{and at } \phi = \pi: \quad T \geq 2(q!)^{1/q}. \quad (71)$$

Moreover (`elementary_gt_four_over_e`) $4(N+1)/e < 2(q!)^{1/q}$ for every $N \geq 0$, so the bound implies, and is strictly stronger than, the closed form $4(N+1)/e$ of Section 4.

Theorem 11.2 (Sharpened bound; `endgame_bound`). *For every $a \geq 2$ and every $N \geq N_0(a)$, with $N_0(a)$ the explicit integer threshold (75), every order- N admissible sequence with $\phi = \pi$ satisfies*

$$4(N+1) \left(1 - a \left(\frac{\log q}{q}\right)^{2/3}\right) \leq T, \quad (72)$$

and the two numerical instances `endgame_bound_2_3_num` and `endgame_bound_2_2_num` bound it explicitly:

$$\begin{aligned} N \geq 5010 &\implies 4(N+1)(1 - 2.3(\log q/q)^{2/3}) \leq T, \\ N \geq 46000 &\implies 4(N+1)(1 - 2.2(\log q/q)^{2/3}) \leq T. \end{aligned} \quad (73)$$

The formal inequality is (72), and it is non-strict: the proof rules out $\tau < \rho_a(q)q$, so strictness is not part of the kernel-checked statement. Combined with the elementary term (`all_N_bound_end`), the same layer gives, for every $N \geq 1$,

$$\max\left(2(q!)^{1/q}, \mathbf{1}_{N \geq N_0(a)} \cdot 4(N+1)(1 - a(\log q/q)^{2/3})\right) \leq T, \quad (74)$$

which is the kernel-checked form of (10).

The threshold is the explicit function `effN0End`, evaluated at the cutoff constant $K_0 = 21.1$ and at $(C_{d1}, C_{d2}, \beta) = (7, 770, 4)$:

$$\begin{aligned} N_0(a) &= \max(\lceil 256a^3 \rceil, \lceil \text{effQ0}/2 \rceil, \lceil \text{effQmin}/2 \rceil), \quad \text{effQ0} = \max(512a^3, (64\text{effK1})^{-1/\text{effE}}), \\ \text{effE} &= \max\left(\beta + \frac{2}{3} - 2s, \frac{3}{2} - \frac{3}{2}s\right), \quad s = \text{effS}(a) = \frac{47}{50}a^{3/2}, \\ \text{effK1} &= \frac{1 + C_{d1}}{36} \left(2704 C_{d2}/a + 5408 K_0 C_{d1}/a + 752 K_0 a^{-3/4}\right), \\ \text{effQmin} &= \max(10020, e, (20e a^{-1/4})^{1/(s/2 - 7/6)}). \end{aligned} \quad (75)$$

The general threshold `effN0` of `Effective.lean` carries the crude floor $5 \cdot 10^9$, which `effN0End` replaces by `effQmin`. The numerical lemmas behind (73) give $-\text{effE}(2.3, 4) \geq 1.85$, $\text{effK1}(2.3) \leq 280249$, $\text{effQ0}(2.3) \leq 10020$, $\text{effQmin}(2.3) \leq 10020$ and hence $N_0(2.3) \leq 5010$; and $-\text{effE}(2.2, 4) \geq 22/15$, $\text{effK1}(2.2) \leq 292954$, $\text{effQ0}(2.2) \leq 92000$, $\text{effQmin}(2.2) \leq 92000$ and hence $N_0(2.2) \leq 46000$ (the source records the sharper value `effN0End` $K_0 2.2 = 45006$ and rounds up).

Table 4: Explicit constants of the machine-checked sharpened bound and the declarations carrying them; the last two rows are the collars instantiated by `collar_bound_concrete`.

Constant	Lean declaration
$\mu(\rho) \geq \frac{47}{50}(1-\rho)^{3/2}$	<code>mu_lower_sharp</code>
$D_0(\rho) \leq 13(1-\rho)^{-1/2}$	<code>D0_le_inv_sqrt</code>
$q_0(C) = \max(10^4, 7 \max(C, 100)^{6/7})$	<code>criterion_param_max</code>
$K_0 = 21.1; \kappa' \leq K_0/B, \kappa'' \leq K_0/B^2$	<code>deriv_cutoff_le_explicit</code>
$\ (\psi_A)'\ \leq (7k^2/\tau + 9)\varepsilon; \ (\psi_A)''\ \leq (384k^4/\tau^2 + 14k^2/\tau + 9)\varepsilon$	<code>norm_deriv(2)..._sharp</code>
$\tau M_1 \leq 7k^2\varepsilon, \tau^2 M_2 \leq 770k^4\varepsilon$	<code>collar_bound_concrete</code>

11.2 The constants

The real-analytic inputs are proved in `RobustZ/EffectiveCalc.lean`: for the contour-shift exponent $\mu(\rho) = \log((1 + \sqrt{1 - \rho^2})/\rho) - \sqrt{1 - \rho^2}$ and its prefactor D_0 , the sharpened layer uses $\mu(\rho) \geq \frac{47}{50}(1 - \rho)^{3/2}$ on $\frac{1}{2} \leq \rho < 1$ and $D_0(\rho) \leq 13(1 - \rho)^{-1/2}$ on $\frac{3}{4} \leq \rho < 1$ (the asymptotic constant $2\sqrt{2}/3$ makes $47/50$ optimal to 0.3%). The criterion $D_0(\varrho_q)e^{-q\mu(\varrho_q)} < \sqrt{\varrho_q}/(Cq)$ packages the two at threshold $q_0(C) = \max(10^4, 7 \max(C, 100)^{6/7})$, $C > 0$; the endgame path uses instead the combined form $D0_mu_le, D_0(\rho_a(q))e^{-q\mu(\rho_a(q))} \leq 13a^{-1/2}(q/\log q)^{1/3}q^{-s}$, with threshold `effQ0`.

The abbreviations `norm_deriv(2)..._sharp` and `..._collar_gen` denote the `sharp` and `gen` instances of `norm_deriv_psiA_scaled_collar_*` and `norm_deriv2_psiA_scaled_collar_*`.

The sharp collar bounds hold on the whole collar $|\mu| \leq \tau + B$ under explicit smallness conditions on B/τ and on the tail ratio r . The generalized forms used by the final assembly (`..._collar_gen`) keep the honest factor $R = 1/(1 - r')$ with $r' = r e^{2\sqrt{B/\tau}}$ instead of assuming it small, $\|(\psi_A)'\| \leq 2e^{2k\sqrt{B/\tau}}(2k^2R + 4R^3 + \tau R)\varepsilon_h/\tau$, with a five-term analogue for $(\psi_A)''$; the exact cancellation

$$\varepsilon_h = e^{\tau \sinh \delta'} e^{-\delta' k} \leq \frac{\varepsilon}{2R} \quad (\text{honest_le_eps_mul}) \quad (76)$$

then removes R and leaves the constants 7 and 770, free of q and of τ . The factor $\frac{1}{36}(1 + C_{d1})$ of (77) is the elementary estimate $(2\pi)^{-2} < 1/36$; at the end of the chain the three terms of (77) are each bounded by `effK1` $q^{\text{effE}}(\log q)^{-1/2}$, and $q \geq \text{effQ0}$ makes their sum smaller than $1/64$. This is where $a \geq 2$ enters: $s = \frac{47}{50}a^{3/2}$ must exceed $\beta/2 + 1/3$, that is $a > 1.834$ for $\beta = 4$; $a = 2$ is a safe margin, not a sharp one.

11.3 Mathematical content of the formalization

Three ingredients are worth naming. First, the doubling of flatness: order- N matrix robustness at $\lambda = 1$ forces $h^{(i)}(1) = 0$ for $i < q = 2N + 2$ (`h_flat`), so the q vanishing derivatives become q vanishing moments (`h_moments`), and `abs_zero_le_of_flat` turns Taylor–Lagrange with q flat terms into $|h(0)| \leq \tau^q/q!$. Second, the collar estimates cover the whole collar $|\mu| \leq \tau + B$ with $B = \sqrt{\varepsilon}$, beyond the band where the polynomial only has to approximate $e^{-i\mu}$; the version with honest $1/(1 - r')$ constants is what makes a width of order $\sqrt{\varepsilon}$ usable with no unproved hypothesis. Third, the product bound is sharpened from the q^8 estimate of Section 5 to

$$C_1 C_2 \leq \frac{1+C_{d1}}{36} (16C_{d2}k^\beta \varepsilon^2 + 32KC_{d1}k^2 \varepsilon^2 + 16K\tau \varepsilon^{3/2}), \quad (77)$$

whose polynomial factor is q^2 rather than q^8 ; the declaration `kernelProd_sqrt_param_nolift` is this statement. The companion shape `kernelProd_sqrt_param` imposes $k^2B \leq \tau, k^4B \leq \tau$

and collapses the two ε^2 terms into a single $\tau\varepsilon^{3/2}$ term with the ε -independent constant $(1 + C_{d1})(16C_{d2} + 32C_{d1}K + 16K)/36$. The scale $B = \sqrt{\varepsilon}$ carries no power of q .

11.4 Scope of the verification

Machine-checked: the elementary bound with its exact constant 2, at every order and every $0 < \phi \leq \pi$; the sharpened bound with its explicit threshold $N_0(a)$ for $\phi = \pi$; and the combined form (74), plus the lemmas they rest on. Not machine-checked: the numerical material of Section 8. The low-order sequences found by local search and their re-verified costs are upper bounds computed outside Lean; they enter no proof, and no optimality or small- N statement about them is formalized. The threshold $N_0(a)$ is conservative by construction: 5010 and 46000 come from rounding constants up and from the crude exponent combination (75), not from the method, and the same mechanism with numerically evaluated constants reaches $a \approx 1.5$ –2 at $q \approx 10^2$. The audit ran on the full development: the build is green (3739 jobs), a source scan finds no `sorry`, `admit`, or `axiom` in the project files, and the eleven headline theorems listed in `RobustZ/AuditFinal.lean` — among them `endgame_bound`, `all_N_bound_end`, `elementary_bound_pi` and `collar_bound_concrete` — depend only on the three standard logical axioms `propext`, `Classical.choice` and `Quot.sound`. Statements, build log and the raw `#print axioms` output are collected at <https://show.dytchem.cn/lean/>, and the repository is cited as [21], with sources at <https://github.com/Dytchem/robust-z-lower-bound-lean>.

12 Reading the bound

The results of this paper are the explicit inequalities of Theorems 2.2 and 2.3 together with the single bound (10) that takes their maximum at each order; the asymptotic coefficient $c_*(\phi) \geq 4$ is a corollary of the second.

The two coefficients. The elementary bound uses flatness alone. Order- N matrix robustness forces the carrier h of (12) to vanish to order $q = 2N + 2$ at $\lambda = 1$ (Proposition 3.1), and with $|h^{(q)}| \leq \tau^q$, $\tau = T/2$, one Taylor–Lagrange step at the endpoint gives the inequality of Theorem 2.2. Its coefficient of N is $4/e \approx 1.4715$, with no threshold in N and no approximation theory; and at $\phi = \pi$ the bound is strictly stronger than the closed form $4(N + 1)/e$ at every order (`elementary_gt_four_over_e`, machine-checked). The sharpened bound uses the whole band instead. The q vanishing derivatives at $\lambda = 1$ are q vanishing moments, into which a polynomial of degree $q - 1$ can be inserted; approximating $e^{-i\mu}$ on the band $|\mu| \leq \tau$ that way, with a compactly supported smoothing, makes $h(0)$ exponentially small in q whenever $\tau \leq \rho q$ with fixed $\rho < 1$, which the endpoint separation (20) forbids. Carrying the constants through gives $T > 4(N + 1)[1 - a(\log q/q)^{2/3}]$ for every $a \geq 2$ and $N \geq N_0(a)$ (Theorem 2.3), a coefficient approaching 4. The two bounds have the same shape, a constant times $(N + 1)$: the factor e between $4/e$ and 4 is what the band supplies over one Taylor remainder.

Where the sharpened term takes over. Table 1 evaluates both at sample orders: the elementary term is the larger one at $N = 5$ and $N = 10$ (10.58 against 4.7; 18.11 against 16.6), the sharpened expression passes it at $N = 12$, i.e. $q = 26$ (22.1 against 21.1), and beyond that their ratio tends to e . Below $N_0(a)$ the sharpened term is switched off in (10), so between $N = 12$ and N_0 the proved bound is still the elementary one: the crossover locates the order from which whole-band information begins to pay, not a point at which the proved coefficient

changes. Both terms grow linearly with the same $(N + 1)$ normalization, so the comparison is one of constants: at least $4(N + 1)/e$ at every order, and $(1 - o(1))4(N + 1)$ at large order.

What the pair rules out. At every order the cost is bounded below by the elementary bound, hence by $4(N + 1)/e$: no arrangement of free rotations, however many, makes high-order robustification sublinear. With the equiangular family, the sharpened bound confines the asymptotic coefficient to

$$4 \leq c_*(\pi) \leq 2\pi, \quad (78)$$

an interval of width $\pi/2 \approx 1.57$; the numerical costs of Section 8 follow $T \approx 5.26N + 2.81$ over $N \leq 8$, within a factor 1.4 of the floor, so what remains to be found is a bounded correction rather than a new regime. And since the model already grants exact, instantaneous, free X rotations in unlimited number and arbitrary placement, and a target reached exactly at the nominal error, the intuition that a rich enough fast-control layer makes the uncertain evolution asymptotically efficient is false here.

Why the idealization makes the bounds conservative. The model is thin by design, and every one of its assumptions favours the designer: one common multiplicative error, no time dependence, no decoherence, no leakage, perfect and instantaneous transverse control. Any model closer to a laboratory — finite X -rotation duration, transverse-control errors, drifting or segment-dependent errors, finite pulse widths, broadband noise — adds constraints, and adding constraints can only increase the least cost of an order- N robust sequence, as long as it is still the accumulation of uncertain Z evolution that is being charged. The bounds are therefore bounds on the best case: a real device cannot do better than its idealization. Nothing is said here about decoherence-limited performance, about errors that are not common to all segments, or about the deviation at one particular ε : shaping the response over an error band, as low-order composite pulses such as BB1 or CORPSE do, is not the same as cancelling derivatives at a point.

Finite order, incoherent errors, and the orders that matter. A device does not get to choose N freely: a longer sequence also spends more time in the presence of decay, the tension that shapes time-optimal gate design on Rydberg platforms [17]. Writing ω for the rate at which the uncertain Z -phase accumulates and γ for the decay rate, an order- N sequence of cost T pays an incoherent infidelity of order $\gamma T/\omega$ while its coherent error falls like ε^{N+1} , and the two balance at

$$N^* \approx \frac{\ln(\omega/\gamma)}{2\ln(1/\varepsilon)}, \quad (79)$$

so the useful order grows only logarithmically as coherence improves. This is where the finite-order form of the bounds matters: the relevant orders are small, and there the operative term of (10) is the elementary one, which supplies the cost side of the balance at every order and without a threshold — each order costs at least $2[(2N + 2)!]^{1/(2N+2)} \geq 4(N + 1)/e$ units of accumulated uncertain phase. The incoherent floor is thus not a property of a particular composite construction, and the tradeoff cannot be evaded by inventing better pulses, only by improving γ , ε , or the resource model itself. Figure 2 evaluates the balance at parameters representative of current neutral-atom arrays: the optimum is at $N^* = 2$, and improving γ/ω by three orders of magnitude moves it only to 4. The panel uses the tabulated costs of Table 2; replacing them by the rigorous floor of (10) lowers the incoherent term but not the coherent one, so the minimum stays at small order. Orders at which the coefficient 4 of Theorem 2.3 becomes operative are thus physically remote, and they recede only logarithmically.

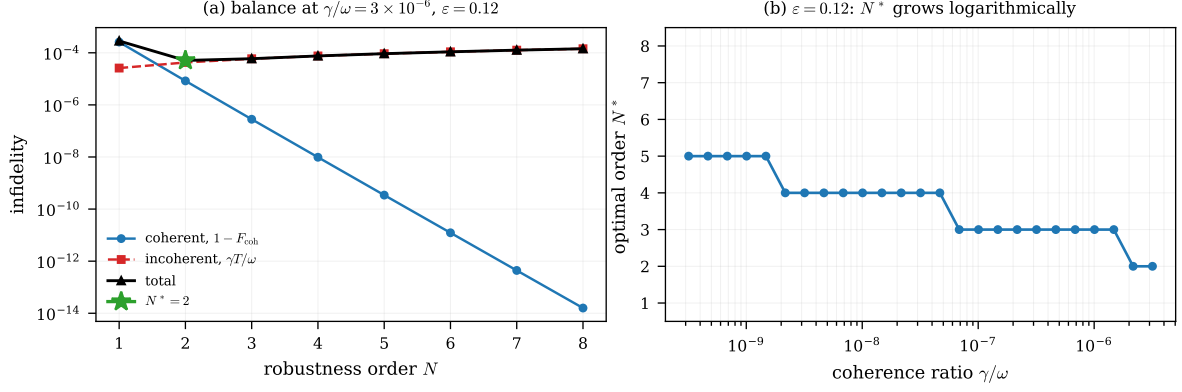


Figure 2: The coherent-versus-incoherent balance at parameters representative of current neutral-atom arrays: interaction strength $V/2\pi \approx 450$ MHz at $2\mu\text{m}$ separation [15] with Rydberg lifetimes of order $10^2\mu\text{s}$ [18], i.e. a coherence ratio $\gamma/\omega \approx 3 \times 10^{-6}$, and a systematic variation $\varepsilon = 0.12$ of the interaction. (a) the coherent infidelity $\sin^2(\delta(\varepsilon)/2)$ of the sequences of Table 2, the incoherent infidelity $\gamma T/\omega$ they accumulate, and the sum; the minimum is at $N^* = 2$. (b) the same balance as the coherence ratio improves: N^* moves only from 2 to 4. The coherent term is measured for the tabulated sequences and is therefore an upper bound on what a better construction of the same order could reach.

13 Discussion

The bounds are lower bounds on a resource, not a pulse-design prescription. Free and exact X rotations can rearrange how the systematic Z error accumulates, but when the common Z scale is turned down to $\lambda = 0$ every costly evolution disappears, and the carrier must connect two separated endpoints while being extremely flat at $\lambda = 1$. What follows is an accounting of the status of each ingredient.

What is machine-checked. The elementary bound is formalized for every order N and every target $0 < \phi \leq \pi$ (`RobustZ.elementary_bound`, with `elementary_bound_pi` the case $\phi = \pi$), together with the pointwise comparison $4(N+1)/e < 2[(2N+2)!]^{1/(2N+2)}$ (`elementary_gt_four_over_e`). The sharpened bound and the combined statement `all_N_bound_end` behind (10) are formalized at $\phi = \pi$: the target angle enters the whole-band argument only through the endpoint separation $h(0) \geq 2\sin^2(\phi/4)$, so the same constants give the general- ϕ form, but the assembled numerical statement in the development is the case $\phi = \pi$ (Section 11). Both explicit constants, including the thresholds, are part of the checked statements.

Thresholds and numerics. The thresholds $N_0(a)$ are sufficient conditions assembled from explicit constants — a cutoff-derivative constant, an approximation rate for $e^{-i\mu}$ on the band, and kernel bookkeeping. The exponent $2/3$ in (9) is the reciprocal of the exponent $3/2$ in that rate: with $1 - \rho = a(\log q/q)^{2/3}$ the exponential gain behaves like $q^{-a^{3/2}}$ against a polynomial loss, so any fixed a above a bound determined by them suffices, and the condition $a \geq 2$ with $N_0(2.3) = 5010$, $N_0(2.2) = 46000$ is conservative bookkeeping rather than a feature of the control problem. The sharpened statement being vacuous below N_0 is therefore not evidence that the coefficient is smaller there; those orders are covered by the elementary bound. Every cost in Table 2, by contrast, is numerical: an upper bound for $T_{\min}(N, \pi)$ from a local search, re-verified independently, with no claim of minimality and no asymptotic content.

The exact finite-order question. It is natural to ask whether the linear bound can be made exact at finite order, in the form $T \geq 4N$ at $\phi = \pi$. No argument of the present type can settle it. The carrier argument uses four properties of h : exponential type $\tau = T/2$, a zero of order $q = 2N + 2$ at $\lambda = 1$, a sup bound, and the endpoint value at $\lambda = 0$. Rescaling λ to unit bandwidth and normalizing the sup bound to one turns an admissible carrier into a real function g of exponential type 1 with $g^{(k)}(0) = 0$ for $k < q$, $|g| \leq 1$ on $\mathbb{R}$, and $|g(-y)| \geq 1/2$ at $y = \tau$; the relaxation of that property list, whose optimum $y_*(q)$ is the least such y , therefore contains every rescaled carrier, and $T \geq 4N$ is exactly $y_*(q) \geq q - 2$. Any proof that uses only those four properties proves a lower bound valid for $y_*(q)$ as well.

That relaxation has been solved numerically, and it does not obey the target. On commensurable frequency grids, with the sup-norm constraint enforced globally on one period, certified by a Bernstein correction, and with the moment residuals at machine precision, explicit exponential sums with $y < q - 2$ exist from $q = 44$, that is $N = 21$, upward; the smallest witness found has $T = 83.44$ against $4N = 84$, and the saving grows with N , reaching about 17% at $N = 29$ ($T = 96$ against 116). The grid restricts the admissible class, so such a solution is a witness for the continuum relaxation as well; the first beating order moves down from $q = 46$ to $q = 44$ when the grid is refined, and none was found below it. The conclusion is negative but useful: the finite-order statement $T \geq 4N$ cannot be obtained from the analytic data of the carrier — band, flatness, sup bound, endpoint value — and any proof of it must use the product structure of (3): that h is the trace overlap $1 - \frac{1}{2} \operatorname{Re} \operatorname{Tr}(U_1^\dagger U_\lambda)$ of an X - Z product with $\theta_j \geq 0$ and $\sum_j \theta_j = T$, so that its frequencies are the half-sums $\frac{1}{2} \sum_j s_j \theta_j$ with $s_j \in \{\pm 1\}$, and its endpoint value is tied to the free rotations through $h(0) = 1 - \cos(\phi/2) \cos(A/2)$ of (19). None of that structure survives the relaxation, whose extremals are edge-dominated packets alternating in parity with q .

The constant is not proved optimal. The whole-band argument proves the left half of (78) and leaves a genuine gap: equality would require a matching upper-bound construction in the precise free- X cost model. Every cost found numerically is strictly below the equiangular cost $(2N + 1)\pi$ of [3] at $\phi = \pi$, by a factor between 0.838 and 0.926, and above the proved floor; since those values are upper bounds from a local search, they neither raise the lower end nor settle the upper end of (78).

14 Conclusion

We have established explicit lower bounds on the Z -evolution cost of high-order robust composite Z rotations, in a model where the imperfect Z evolution is the only costly operation and exact X rotations are free. For every order $N \geq 1$ and every target $0 < \phi \leq \pi$ the elementary bound gives $T \geq 2[2(2N + 2)! \sin^2(\phi/4)]^{1/(2N+2)}$, a coefficient $4/e \approx 1.4715$ that is pointwise stronger than $4(N + 1)/e$. Using the whole frequency band of the carrier as well gives, at $\phi = \pi$, $T > 4(N + 1)[1 - a(\log q/q)^{2/3}]$ for every $a \geq 2$ and every $N \geq N_0(a)$, with $N_0(2.3) = 5010$, so that the coefficient approaches 4. Taking the larger of the two at each order yields a single bound valid at every order (Corollary 2.4), of which the familiar asymptotic statement $\liminf_{N \rightarrow \infty} T_{\min}(N, \phi)/N \geq 4$ is a corollary. Both constants, and the thresholds that come with them, are machine-checked in Lean 4 with Mathlib.

What remains open is a sharper finite-order statement. The numerical constructions of Section 8 are admissible sequences whose nominal identity and order of robustness were re-verified independently, and they show that the equiangular value 2π is not optimal, by a factor between 0.838 and 0.926; being upper bounds from a local search, they determine no value of $c_*(\pi)$. In the other direction, the relaxation discussed above shows that the exact

statement $T \geq 4N$ cannot be reached from the analytic data of the carrier, since certified band-limited constructions with $T < 4N$ exist from $N = 21$ on. Closing the gap between $4/e$ and 4, or proving a bound exact at finite order, therefore requires an argument that uses the product structure of (3) rather than the carrier alone; at the orders a device actually uses, the elementary bound is the one that applies.

A Elementary derivative estimate

For completeness, consider the j th costly factor

$$Z_j(\lambda) = \exp\left(-i\frac{\lambda\theta_j}{2}\sigma_z\right).$$

Because the generator has operator norm $1/2$,

$$Z_j^{(m)}(\lambda) = \left(-i\frac{\theta_j}{2}\sigma_z\right)^m Z_j(\lambda), \quad \|Z_j^{(m)}(\lambda)\| = \left(\frac{\theta_j}{2}\right)^m.$$

Differentiate the product U_λ . In a term corresponding to a composition $k_1 + \dots + k_L = k$, all undifferentiated factors are unitary, while the differentiated factors contribute exactly the product of the above norms. Hence

$$\|U_\lambda^{(k)}\| \leq \sum_{k_1 + \dots + k_L = k} \frac{k!}{k_1! \dots k_L!} \prod_j \left(\frac{\theta_j}{2}\right)^{k_j}.$$

The multinomial theorem gives τ^k . Finally,

$$|h^{(k)}(\lambda)| \leq \frac{1}{2} |\text{Tr}(U_1^\dagger U_\lambda^{(k)})| \leq \|U_\lambda^{(k)}\| \leq \tau^k.$$

B Why matrix flatness doubles for the carrier

Let $V_\lambda = U_1^\dagger U_\lambda \in SU(2)$. Every element of $SU(2)$ can be written as

$$V_\lambda = a(\lambda)I + i\mathbf{b}(\lambda) \cdot \boldsymbol{\sigma}, \quad a^2 + |\mathbf{b}|^2 = 1,$$

with real a and $\mathbf{b}$. Since $U_1 = U_\lambda$ at $\lambda = 1$, $a(1) = 1$ and $\mathbf{b}(1) = 0$. The order- N matrix flatness means $V_\lambda - I = O((\lambda - 1)^{N+1})$, hence

$$\mathbf{b}(\lambda) = O((\lambda - 1)^{N+1}), \quad 1 - a(\lambda) = O((\lambda - 1)^{2N+2}).$$

But $h = 1 - a$, proving order $2N + 2$. This is the exact analytic reason for the factor of two in the $4/e$ estimate.

C The smoothing L^1 estimate

The proof of (42) is included because it is the simplest analytic component of the formalized whole-band argument. From (39),

$$|\kappa(t)| \leq C_1.$$

Twice integrating by parts gives

$$\kappa(t) = \frac{-1}{2\pi t^2} \int_{\mathbb{R}} \hat{\kappa}''(\mu) e^{-i\mu t} d\mu,$$

so $|\kappa(t)| \leq C_2/t^2$. Put $t_0 = \sqrt{C_2/C_1}$ and integrate separately over $|t| \leq t_0$ and $|t| > t_0$:

$$\begin{aligned} \|\kappa\|_{L^1} &\leq 2C_1 t_0 + 2 \int_{t_0}^{\infty} \frac{C_2}{t^2} dt \\ &= 2\sqrt{C_1 C_2} + 2\sqrt{C_1 C_2} = 4\sqrt{C_1 C_2}. \end{aligned}$$

Since $|h(t)| \leq 2$, $|\int h\kappa| \leq 2\|\kappa\|_{L^1}$, which gives (45).

D Contour-shift estimate for the approximation coefficients

Let

$$f(z) = e^{-i\tau \cos z} e^{-inz}.$$

The function is entire and 2π -periodic. By periodicity and Cauchy's theorem, the real-line integral defining c_n can be shifted to $z = u - i\delta$ for $\delta > 0$:

$$c_n = \frac{1}{2\pi} \int_0^{2\pi} f(u - i\delta) du.$$

The shifted modulus obeys

$$|e^{-in(u-i\delta)}| = e^{-n\delta},$$

and, because $\cos(u - i\delta) = \cos u \cosh \delta + i \sin u \sinh \delta$,

$$\left| e^{-i\tau \cos(u-i\delta)} \right| = \exp(\tau \Im \cos(u - i\delta)) = \exp(\tau \sin u \sinh \delta) \leq e^{\tau \sinh \delta}.$$

Therefore

$$|c_n| \leq e^{\tau \sinh \delta - n\delta},$$

which is (49). Summing the geometric tail gives the exponential form (50) used in the main proof.

E The collar estimate behind the product bound

This appendix records what enters (58). The constants are those of the formal development, which proves the inequality under the side conditions $\tau \leq \rho q$, $\beta = B/\tau = \eta/q^2 \leq 1/2$ and $\delta > 2\sqrt{\beta}$.

(a) *Chebyshev collar bounds.* The only delicate point in differentiating the polynomial approximation outside the exact band is the second derivative of T_n . Put $u = \cosh y$, so that $T_n(\cosh y) = \cosh ny$ and $T'_n(\cosh y) = n \sinh(ny)/\sinh y$; the second derivative is singular in this parametrisation at $y = 0$, and the standard identity

$$(1 - u^2)T''_n(u) = uT'_n(u) - n^2 T_n(u)$$

is what removes the singularity: at $u = 1$ it gives $T''_n(1) = n^2(n^2 - 1)/3$. Combining the two descriptions — the explicit hyperbolic combination on the collar and the differential equation at its boundary — gives, for $|u| \leq 1 + \beta$ with $0 \leq \beta \leq \frac{1}{2}$ and $y \leq \operatorname{arcosh}(1 + \beta) \leq \sqrt{2\beta}(1 + O(\beta))$,

$$|T_n(u)|, |T'_n(u)|, |T''_n(u)| \leq 32n^4 e^{2n\sqrt{\beta}},$$

the collar growth of Chebyshev polynomials and their derivatives [19].

(b) *Assembly*. The two constants of (41) are integrals over the support $[-(\tau + B), \tau + B]$, so each is bounded by the length of the support times the supremum of its integrand. On the band $\hat{\kappa} = \Psi$ and $|\Psi| \leq \eta$; on the collar $\hat{\kappa} = \Psi\chi$, and the product rule with (38) gives

$$|\hat{\kappa}''| \leq |\Psi''| + \frac{2K}{B}|\Psi'| + \frac{K}{B^2}|\Psi|.$$

The polynomial P_{q-1} and its first two derivatives are controlled on the whole collar by (a), with the head bound $|c_n(\tau)| \leq 1$ for $n < q$ and the tail (49) for $n \geq q$. Running the explicit bookkeeping with $\beta = \eta/q^2$ and $\eta/B = q^2/\tau$ gives $BM_1 \leq 9\eta q$ and $\tau^2 M_2 \leq 141q^5$, where M_1 and M_2 bound $|\hat{\kappa}|$ and $|\hat{\kappa}''|$ on the support; multiplying yields (58) and, with (53), (59). This is exactly the place where the relative collar scaling $B/\tau = \eta/q^2$ is used.

F Formalization versus elementary exposition

Three distinctions are worth keeping explicit.

First, the elementary bound of Section 4 is not merely expository: it is a machine-checked theorem (`elementary_bound`, `elementary_bound_pi`), together with the pointwise comparison `elementary_gt_four_over_e` with $4(N+1)/e$. Only the weaker $2/e$ estimate of Section 4 remains paper-level; it is superseded by the doubled-flatness bound and is kept to show how little is needed for linear growth.

Second, the present proof description uses the language of Chebyshev approximation because it makes the whole-band idea transparent. The formal development realises the approximation in two ways: the band error through the Fourier coefficients of $e^{-i\tau \cos u}$ and contour shifting (`Approx.lean`, `Band.lean`), and the collar derivative bounds of (65) through the Chebyshev tail of the same expansion (`SharpDeriv.lean`, `SharpDerivGen.lean`). Neither route relies on a separate Bernstein-ellipse theorem as a black box.

Third, the asymptotic theorem in `Theorem.lean` carries the two representation-level assumptions discussed in Section 11, because it is phrased through Mathlib’s real-valued `liminf`; the underlying analytic contradiction is the unconditional statement that an admissible sequence cannot satisfy $\tau \leq \rho(2N+2)$ for all sufficiently large N . The finite-order theorems of Section 11 need no such hypothesis: they are statements about the cost of an admissible sequence and carry nothing beyond admissibility.

G Numerical solutions

Table 2 is reproduced here as explicit data. For every order $N \leq 12$ at $\phi = \pi$ we list the L costly angles θ_j and the L free angles α_j in units of π , both in the *time order* of the sequence: α_j is the free rotation that follows the costly segment θ_j , and the first printed entries are the ones applied first. The product that was verified is therefore

$$U_\lambda = X(\alpha_L)Z(\lambda\theta_L)X(\alpha_{L-1})Z(\lambda\theta_{L-1})\cdots X(\alpha_1)Z(\lambda\theta_1),$$

which is (3) with the indices relabelled, $j \mapsto L+1-j$. Two conventions turn the raw optimiser output into these lists, and both are exact rather than cosmetic: surplus free rotations, where the search produced $L+1$ of them, are merged with $X(a)X(b) = X(a+b)$, and every α_j is reduced to its representative in $[0, 2\pi)$, which uses $X(\alpha+2\pi) = -X(\alpha)$ and therefore changes the propagator only by the global sign that the convention $U_1 = Z(\pi)$ of (4) restores. After those two steps each row has L angles of each kind, as the sequence of the paper requires.

Each printed list was re-verified from scratch in exactly this form, as described in Section 8: the nominal mismatch $\|U_1 \mp Z(\pi)\|$ is below 9×10^{-10} and the flatness residual ρ_N of (23) is below 1.5×10^{-10} , so the printed lists carry the accuracy claimed in Table 2; the deviations from the target follow ε^{N+1} accordingly. The same lists, with the Bloch-sphere trajectory of every sequence and a browser-side re-computation of these numbers, are the first resource of Table 3.

$N = 1$ ($L = 17$, $T = 8.1505$, $T/N = 8.15$):

θ/π (time order): +0.108125183599, +0.092383883307, +0.088416321466, +0.092554145344, +0.108753457496
+0.373020695718, +0.111493518906, +0.095912685220, +0.082164186029, +0.085272886168
+0.110760945924, +0.393902410229, +0.123722301608, +0.107509875814, +0.107603543063
+0.124084388415, +0.388698205781
 α/π (time order): +0.139509498938, +0.145944348983, +0.145870810709, +0.139250653694, +0.123533515476
+1.873548764127, +1.855017527164, +1.849452085074, +1.877288617724, +1.855962340447
+1.872412643449, +0.151026665920, +0.169473165152, +0.174726658455, +0.169331711689
+0.149513806909, +1.877132330900

$N = 2$ ($L = 21$, $T = 13.3466$, $T/N = 6.67$):

θ/π (time order): +0.144180195202, +0.807347377618, +0.111301558678, +0.090600304171, +0.086924269216
+0.097056360512, +0.406097959760, +0.164364922580, +0.187385727145, +0.511527875184
+0.227796958327, +0.144432089731, +0.135039100093, +0.000000000000, +0.368242305408
+0.098114729953, +0.083610128733, +0.000000000000, +0.084306168202, +0.100828811037
+0.399195860623
 α/π (time order): +0.101628388503, +0.166188820243, +0.194802631072, +0.205084531062, +0.201890059349
+0.181516421378, +1.724482705429, +1.714856535448, +1.758458401688, +1.909778799575
+1.854338564622, +1.838884382710, +1.400055433759, +0.444259769013, +0.189952413800
+0.216864606970, +0.447473445912, +1.776599612876, +0.215208004840, +0.191213984707
+1.885525144298

$N = 3$ ($L = 25$, $T = 18.5505$, $T/N = 6.18$):

θ/π (time order): +0.132620428470, +0.131467794724, +0.160912920764, +0.295403035709, +0.165046632787
+0.099074314397, +0.079205690164, +0.072230594659, +0.068744196277, +0.075041880203
+0.297845621053, +0.097397497973, +0.088307245791, +0.097433138552, +0.140648244647
+0.579912272731, +0.242596023047, +1.019217761231, +0.378279813873, +0.097171679919
+0.083570562454, +0.093455518886, +0.353878883662, +0.141465636776, +0.913868789512
 α/π (time order): +1.871362772045, +1.876745037440, +1.897323737294, +1.918050508467, +1.876098599105
+1.853586780782, +1.840899074031, +1.838061118302, +1.857410725610, +1.861457293321
+0.209566228343, +0.232060044895, +0.231759451612, +0.210594949975, +0.156457878316
+0.199750509404, +0.215022749003, +0.116601098108, +1.738297095933, +1.698357014217
+1.695810105036, +1.729832569786, +0.182666468938, +0.175022029574, +0.121487482170

$N = 4$ ($L = 31$, $T = 23.7878$, $T/N = 5.95$):

θ/π (time order): +0.436888342592, +0.470606581399, +0.129518418028, +0.113936400796, +0.119391671961
+0.154443561388, +0.151853151655, +0.171032689745, +0.105046693769, +0.083867014309
+0.078892304920, +0.085682265375, +0.000000000000, +0.283856364374, +0.087420993977
+0.080787770445, +0.088820326552, +0.122084601114, +0.609442764306, +0.291981231347
+0.583784556986, +0.240257453646, +0.202109213113, +0.869917532651, +0.364995894510
+0.079868192284, +0.069395619075, +0.078020030268, +0.371046261304, +0.528250310546
+0.518686592092
 α/π (time order): +0.082416868876, +1.867912526602, +1.849709969665, +1.845852316376, +1.850515162200
+1.894347123209, +1.902156646613, +1.840517978476, +1.813497954381, +1.798835844601

+1.799322776273, +1.539027396538, +0.282218130761, +0.207471005384, +0.232833633816
+0.238194236728, +0.209728325420, +0.151102419680, +0.182284212053, +0.167997266240
+0.164976627288, +0.208922402286, +0.202116201061, +0.183718667466, +1.728056807748
+1.689560325197, +1.688286688235, +1.723895093022, +0.239584344067, +1.865360787033
+0.037224409633

$N = 5$ ($L = 38$, $T = 29.0593$, $T/N = 5.81$):

θ/π (time order): +0.357991811292, +0.179208277471, +0.181894250444, +0.255938464975,
+0.296583914430
+0.172686981747, +0.152416407867, +0.180154845256, +0.337005407506, +0.127085880100
+0.087607363662, +0.077875935910, +0.083489193503, +0.274732194353, +0.000000000000
+0.000000000000, +0.085222612003, +0.079456241872, +0.090320841657, +0.135254213530
+0.276909196058, +0.174187347696, +0.145987798161, +0.168240642556, +0.000000000000
+0.681408994806, +0.191886202410, +0.183869911366, +0.833315271263, +0.550413470569
+0.756817851434, +0.063975592300, +0.055762569232, +0.061588495396, +0.354055653573
+0.892752038893, +0.171738307217, +0.532040522200
 α/π (time order): +1.887480536822, +1.874370161668, +1.887263263470, +1.907627140168,
+1.860250181928
+1.840494369988, +1.839780712199, +1.866112704259, +1.846998694357, +1.793594508946
+1.769529299791, +1.767007424800, +1.790198305684, +1.184339720684, +1.713089211853
+1.327382662785, +0.249845316005, +0.246197582801, +0.217455839748, +0.153344232518
+0.150288624619, +0.189834670059, +0.189873235251, +1.508417416171, +0.651004806793
+0.104065658553, +0.120134555302, +0.111563539068, +0.074088799091, +1.819149426067
+1.726261078160, +1.689826467352, +1.687136964194, +1.718428578163, +0.370589059115
+0.140719340831, +0.138098399278, +1.975377212508

$N = 6$ ($L = 48$, $T = 34.2512$, $T/N = 5.71$):

θ/π (time order): +0.782349475959, +0.533916802169, +0.135052524844, +0.120894281671,
+0.138639542091
+0.175927949215, +0.272662445618, +0.208423614150, +0.160080641910, +0.180565710443
+0.343810380815, +0.124101497000, +0.083227811182, +0.073893011140, +0.080850853267
+0.239834122727, +0.000000000000, +0.000000000000, +0.069181852736, +0.063704981259
+0.063614978037, +0.069848220921, +0.091130273942, +0.126276739084, +0.120614535984
+0.149392690127, +0.116707772626, +0.114602363916, +0.139735017751, +0.000000000000
+0.848505799320, +0.538518813165, +0.269066126440, +0.175807990057, +0.172743585476
+0.510804429899, +0.890843656126, +0.670830300632, +0.055027353806, +0.044961377074
+0.044352505600, +0.052683807142, +0.242159614909, +0.113183777281, +0.748681602203
+0.124190227848, +0.134790192691, +0.486288569186
 α/π (time order): +0.090863410439, +1.867990714674, +1.865372807903, +1.847308888258,
+1.856197712735
+1.882974930561, +1.879850326484, +1.837201081727, +1.827588592437, +1.848417144664
+1.844763117637, +1.783673509183, +1.760210585436, +1.758126853329, +1.780060676169
+1.163945183947, +1.692950660456, +1.307805749335, +0.206061632509, +0.210039181383
+0.193587414674, +0.160147698393, +0.115135288859, +0.076213855786, +0.103011262828
+0.171843555289, +0.174961389420, +0.165134835007, +1.494764104385, +0.637071587166
+0.140531814158, +0.100130565002, +0.154154157435, +0.167853979437, +0.156339102255
+1.920836163422, +1.761628776992, +1.779463394147, +1.743250961554, +1.731261556678
+1.739961889829, +1.773287424272, +0.220778332941, +0.210127358332, +0.110277212561
+0.112970857902, +0.101962690610, +1.953965051811

$N = 7$ ($L = 55$, $T = 39.7030$, $T/N = 5.67$):

θ/π (time order): +0.000000000000, +0.382502978561, +0.215113009117, +0.250022283376,
+0.578640334595
+0.246168113124, +0.245964853715, +0.341404678639, +0.206383885863, +0.154920585844
+0.170812263759, +0.250754239841, +0.121501525084, +0.082139971969, +0.071790784627
+0.079529802387, +0.210753260177, +0.000000000000, +0.000000000000, +0.067859490483
+0.058938178783, +0.058478302960, +0.063248636765, +0.084704609756, +0.134782912304
+0.135703584944, +0.102421482199, +0.108812345525, +0.113153938003, +0.134736241373
+0.205186479110, +0.000000000000, +0.370059039446, +0.000000000022, +0.232202767333
+0.248160807430, +0.572190130513, +0.286836413978, +1.037048363405, +0.116402622057
+0.943306321295, +0.774062774200, +0.352168405497, +0.103907898604, +0.488368448347
+0.140030920929, +0.123334193727, +0.000000000000, +0.000000000000, +0.000000000000

+0.000000000000, +0.374336016971, +0.485748353019, +0.574931130110, +0.538330156854
 α/π (time order): +0.493779289902, +1.900889569070, +1.876336350757, +1.908317558970,
+1.837146011092
+1.816152080430, +1.843099627914, +1.834683911942, +1.785122664986, +1.785110791198
+1.813151935125, +1.808730702711, +1.770107666144, +1.749446125451, +1.746670710366
+1.786488722343, +1.164677556157, +1.691766055236, +1.306348247212, +0.209896352729
+0.214702190927, +0.197478795552, +0.163891682090, +0.108684962955, +0.051650421933
+0.093695291191, +0.083567867243, +0.189346350426, +0.182652646879, +0.165060211621
+1.478262577242, +0.618591411940, +0.130624530403, +0.033341235426, +0.210472169768
+0.168207064250, +0.130773110300, +0.151635378939, +0.122226674967, +0.017589741823
+0.118960409759, +0.391439500345, +1.386531781189, +1.433305021050, +1.919957749542
+1.867780452476, +1.280786780884, +1.995690618550, +1.815762734939, +0.468325194475
+0.347354596874, +0.136812697740, +1.898157280104, +0.152321393509, +0.530930909728

$N = 8$ ($L = 58$, $T = 44.9604$, $T/N = 5.62$):

θ/π (time order): +0.000000000000, +0.000000000003, +0.199349404906, +0.236227188583,
+0.570243350542
+0.187989234702, +0.000000000000, +0.151494448355, +0.165093339751, +0.241532618020
+0.215347091675, +0.143929226688, +0.129037241987, +0.151206999300, +0.211124117128
+0.110725316282, +0.070621191381, +0.058700664798, +0.055089698476, +0.061643609511
+0.206803358639, +0.000000000000, +0.000000000000, +0.063099239953, +0.056097624098
+0.057698216330, +0.068706717817, +0.103085111902, +0.184963126694, +0.151243020566
+0.104793414373, +0.103577833541, +0.126244311030, +0.225186491250, +0.387396288528
+0.000000000000, +0.241584293157, +0.326223005821, +0.409077107269, +0.309243546028
+0.698211594800, +0.278790979313, +1.097688072001, +0.999911240421, +0.727454701448
+0.325952520906, +0.091590498083, +0.465768956294, +0.121538522735, +0.080657435359
+0.000000000000, +0.000000000000, +0.000000000000, +0.381873349158, +0.498643891132
+1.510285192266, +0.171809749767, +0.776781914041
 α/π (time order): +0.069448863512, +0.625263370109, +1.867562668693, +1.877858404481,
+1.860275577907
+0.267700690936, +1.559542976207, +1.822104717348, +1.838086922938, +1.874109410942
+1.842218592546, +1.819838219410, +1.817438244716, +1.853474575185, +1.852600967653
+1.812272882271, +1.790176615266, +1.776183707302, +1.797540409109, +1.810026759077
+1.162907227506, +1.693082268757, +1.322484987388, +0.208358727455, +0.212161576036
+0.198839012916, +0.171180638966, +0.116419302815, +0.117769814020, +0.179723762410
+0.188693455102, +0.193310307505, +0.166938515843, +0.104580544571, +1.528573596942
+0.654595113310, +0.182546965542, +0.131727179641, +0.172842939680, +0.161689212174
+0.120731341187, +0.143754447715, +0.229454382523, +0.103461271906, +0.389271862266
+1.385094926946, +1.432787711358, +1.918719346631, +1.865204472486, +1.268428182936
+1.819500290279, +0.469950886040, +0.348312858189, +0.129133220496, +1.864644546112
+0.078500256410, +0.060691711074, +0.805965047218

$N = 9$ ($L = 63$, $T = 52.2657$, $T/N = 5.81$):

θ/π (time order): +0.000000000027, +0.000000000031, +0.030441195677, +0.302217700345,
+0.359298230613
+0.581781106161, +0.264798658303, +0.000823467236, +0.136974563957, +0.145198481232
+0.203482732767, +0.108308671875, +0.158968403341, +0.108733988813, +0.101337274921
+0.134589316483, +0.192332890177, +0.092535128900, +0.055210217906, +0.050810104133
+0.051253742538, +0.059739264487, +0.191956624343, +0.000000000002, +0.000000000003
+0.062448956008, +0.054935469777, +0.056260261051, +0.070094310680, +0.103348930104
+0.175052917943, +0.140941287200, +0.091719911700, +0.079289058310, +0.114431987915
+0.259076009605, +0.396300998279, +0.000013634716, +0.293239057177, +0.522982977034
+0.475046414682, +0.490710549124, +0.537796941860, +0.647927214581, +0.662812083249
+1.145543527128, +0.883866926467, +0.653726653250, +0.290366601311, +0.080441440110
+0.412685431213, +0.106941057017, +0.084586897361, +0.000000000000, +0.000000000000
+0.000459232184, +0.342127148641, +0.156073971459, +0.481761475900, +2.110880208477
+0.174052805124, +0.457725726588, +0.690221374394
 α/π (time order): +0.176172960056, +0.152321529467, +0.301464056310, +1.943496635890,
+1.710939676065
+1.905508946614, +0.249177758892, +1.542789729843, +1.811723134467, +1.820089192277
+1.833332054268, +1.941332095962, +1.842640680347, +1.841832846537, +1.843294043445

+1.869794218606, +1.850578438761, +1.810342071938, +1.790522022131, +1.779740179024
+1.806695537013, +1.839474552446, +1.166542477693, +1.696304085852, +1.328278586755
+0.214339052436, +0.218526652189, +0.205099213313, +0.174128657960, +0.111407970281
+0.100480356875, +0.187394562055, +0.200534356318, +0.202103130421, +0.157456326787
+0.065009030483, +1.570074003283, +0.649190980315, +0.272838812010, +0.261685205226
+0.289846415771, +0.395781107657, +1.824796577150, +1.702764126497, +0.226086530016
+0.228537589517, +0.107922577159, +0.388354736398, +1.383946257584, +1.431922901174
+1.917750595675, +1.864870264968, +1.268957430410, +1.819987660750, +0.470364732684
+0.348715251999, +0.130806629008, +1.930684799213, +1.847401127755, +0.558381846594
+1.980629012294, +0.200978288966, +0.429916020776

$N = 10$ ($L = 70$, $T = 56.9380$, $T/N = 5.69$):

θ/π (time order): +0.135785183283, +0.354287957204, +0.098428313698, +0.094468249186,
+0.325505344031

+0.123653166022, +0.312034466551, +0.094762991590, +0.384916822676, +0.149769891900
+0.146976953677, +0.205217444913, +0.498694938147, +0.317194466750, +0.197694702163
+0.178282679519, +0.241067617803, +0.357318709009, +0.369036865733, +0.059992674918
+0.151209581492, +0.133559274835, +0.101155768142, +0.379581700660, +0.719179334369
+0.295074957781, +0.501973390818, +0.513410998730, +0.232697367264, +0.232734499277
+0.469449277333, +0.265722918417, +0.265723834862, +0.408089765240, +0.176805773792
+0.176622197807, +0.170977138480, +0.170977642675, +0.353966800773, +0.222880108413
+0.222891073807, +0.194356118296, +0.128315863469, +0.260121508695, +0.128590838308
+0.138331320639, +0.317521302804, +0.154612557849, +0.155253144379, +0.508366460168
+0.117768492280, +0.151302384316, +0.244558825556, +0.243370104462, +0.213643377597
+0.160756650937, +0.159652308467, +0.141119021106, +0.136932031161, +0.189007171099
+0.194177746085, +0.180449032668, +0.180524941553, +0.691059114806, +0.596448988323
+0.531364659253, +0.281542600499, +0.404687972626, +0.307014099794, +0.203307400419

α/π (time order): +0.087088987529, +1.617798207047, +1.573870295829, +1.609547784075,
+0.195134421655

+0.181187479290, +1.954525241547, +1.936529583610, +0.196861810020, +0.211816257870
+0.193344112441, +0.148467882953, +0.166902278471, +0.152288788167, +0.066603442918
+0.135950903571, +0.169029354936, +1.987784890100, +0.099776101512, +0.149428141034
+0.659995668547, +0.090360976519, +0.046876576751, +1.550396971908, +1.599877693547
+1.550594988051, +1.969633196967, +1.883339870401, +1.995351686775, +1.995780181712
+0.114854560944, +1.999870794264, +0.127783963878, +0.704616094605, +0.215201717821
+1.973342897555, +1.999036691489, +1.267683643829, +1.702692208953, +1.981621072594
+1.530924132857, +1.810557162165, +1.866625497009, +0.029852253183, +0.037945276043
+0.108679556716, +1.960592380422, +1.977955027163, +1.956381680712, +0.037870628948
+1.971368401853, +0.054043700707, +0.013609035003, +1.905937147433, +1.424928773861
+1.898529261845, +0.010922128913, +0.215297726338, +0.515436211267, +0.277910411055
+0.216522323712, +0.038148449677, +0.457823698721, +1.676227621329, +0.034125683065
+1.926928752696, +1.968557497197, +0.055143703397, +0.035408016159, +1.941447632943

$N = 11$ ($L = 75$, $T = 66.9068$, $T/N = 6.08$):

θ/π (time order): +0.174407772943, +0.146428975925, +0.198226283621, +0.175492805738,
+0.238798900318

+0.128641438592, +0.129051370928, +0.239498357113, +0.236134543578, +0.230404340270
+0.226175783965, +0.241043454149, +0.242134607055, +0.242076171205, +0.241971739605
+0.241202965144, +0.239311577344, +0.218229680693, +0.218961295415, +0.217968992120
+0.218362263209, +0.231833316209, +0.227024788011, +0.458524765943, +0.257467436203
+0.257734954826, +0.252192119555, +0.129071722406, +0.133180767995, +0.214331737424
+0.218366364236, +0.377713499338, +0.456961174336, +0.278817830598, +0.277311556724
+0.471273687160, +0.471874179754, +0.252159669102, +0.243984216581, +0.474240919656
+0.220595566210, +0.220745237549, +0.446041257337, +0.464290507785, +0.466652691260
+0.878265354752, +0.473423307029, +0.244353391714, +0.130170216154, +0.129890624673
+0.934752510614, +0.221462260464, +0.216320086862, +0.222938180695, +0.221784112989
+0.241926407105, +0.242299866444, +0.123834290065, +0.123833012965, +0.242247612945
+0.238345264862, +0.238881298654, +0.122777511087, +0.121739787793, +0.238138274845
+0.452548697173, +0.224304674397, +0.224267843557, +0.475292243796, +0.246473723644
+0.245983577380, +0.487498919413, +0.483707436385, +0.444841113030, +0.457885639586

α/π (time order): +0.184570082485, +0.110445861974, +1.924638849815, +1.461347975717,
+1.952356589093

+1.996416318826, +1.974134592638, +1.925487101736, +1.311212223293, +1.876857668462
+1.917727184761, +1.989747888279, +1.734555617814, +1.994478730573, +1.989598855761
+1.977145604245, +1.448118532530, +1.975276912442, +0.056973696719, +0.009594218167
+0.768597259065, +0.019274532014, +0.072852716558, +0.163849321152, +1.994172040935
+1.965518943660, +1.912608499839, +1.972642626554, +1.401669807820, +1.950632831894
+0.120381619977, +0.786279116052, +0.114921772743, +1.987095008066, +0.108627571713
+0.007126071862, +1.980520880496, +0.017723911094, +0.019067470819, +1.626314955473
+1.996220665760, +0.028439755290, +0.960839952791, +0.024822301070, +0.300661378485
+1.542585628233, +1.964120951387, +1.989561282384, +0.007417456943, +1.789443507049
+1.146692844883, +1.944491453080, +1.923994463130, +0.022505903921, +0.359518877717
+0.008624788741, +1.997320357872, +0.000425762667, +0.004646993280, +0.214413452999
+0.010189585953, +1.987191144369, +1.980798839919, +1.986931244739, +1.138571028390
+1.974069745719, +0.004553050128, +0.523655693661, +0.014735901306, +0.001597692161
+0.151131282197, +0.029862173347, +1.959656834714, +1.844060397783, +1.256394075000

$N = 12$ ($L = 78$, $T = 67.6302$, $T/N = 5.64$):

θ/π (time order): +0.869220884187, +0.291032995856, +0.291937515126, +0.203887232094,
+0.313468227027

+0.475070637312, +0.486473784419, +0.403852827388, +0.349180645610, +0.420239282667
+0.254905752463, +0.254873728160, +0.164696589629, +0.152282383030, +0.396596115289
+0.233689009201, +0.222498457027, +0.202581225945, +0.128151008351, +0.106134872809
+0.046160135901, +0.154906995625, +0.154435539935, +0.504284315515, +0.285462478580
+0.292198171012, +0.266465585459, +0.258054603483, +0.137958416154, +0.137714134039
+0.266393947250, +0.406140077692, +0.310302166125, +0.220730803156, +0.235610669494
+0.251337740361, +0.252534937684, +0.340700766091, +0.312576065922, +0.354261835033
+0.524887208849, +0.219663044133, +0.204589050944, +0.452156979765, +0.417275416571
+0.657067715055, +0.152362233412, +0.131697951063, +0.132322337880, +0.421132209467
+0.215450001750, +0.489254090222, +0.292350565571, +0.152714438559, +0.155704396152
+0.465378895275, +0.154293725603, +0.120988639342, +0.305568731245, +0.084324962978
+0.099040904671, +0.149634813164, +0.348555691249, +0.210899479393, +0.205758741046
+0.493608606321, +0.136871095919, +0.137016018040, +0.676681401416, +0.155545876482
+0.132009931621, +0.321513889986, +0.161014647043, +0.167880008744, +0.301026437474
+0.309859711646, +0.154702999401, +0.207562848166

α/π (time order): +0.180577404807, +0.005680640653, +1.856832128231, +1.908284947577,
+1.939269707489

+1.980697389009, +0.283683646218, +1.871917396571, +1.210261762526, +1.931039640288
+0.005030065143, +1.913992356418, +1.920917139540, +1.916519510348, +0.238035752670
+1.984976447902, +0.007406418866, +0.183674314201, +0.736625225827, +0.176444559531
+0.086338583570, +1.992567186687, +1.590668065830, +1.997522608144, +1.931233657551
+1.678610617261, +1.946048009632, +1.972736566527, +0.008703338486, +0.020069169114
+0.366523633248, +1.772749638498, +1.308801336414, +1.862895621116, +1.959890097711
+1.991447986817, +1.703603768887, +0.038778857595, +0.422784745544, +0.149960846963
+0.222564350404, +0.221301166143, +0.150036218756, +0.045948358862, +0.076951154225
+0.136395882882, +0.230722982445, +0.301147226694, +0.186894499854, +1.682209033492
+1.676146213260, +1.845213814777, +1.691471375776, +1.815140196017, +1.745773350328
+1.710267867242, +1.602901415656, +1.685126191119, +0.260097919883, +0.306076649012
+0.367030516007, +0.237883940369, +0.210822680127, +0.235650300088, +0.194035692081
+1.955279138745, +1.979173483068, +1.889495270102, +0.361042593180, +0.269703827318
+0.290688398423, +1.924555409993, +1.769429671561, +1.841954231661, +1.949016035420
+1.768028584255, +1.62206491417, +1.670730393244

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